

# Reconfigurable Homomorphic Computing

A Living Textbook

Netlist-Clocked Torus FHE · Differentiable Hardware · Adversarial Polymorphic Ciphers

Digital Defiance / HELUT Project

Living edition 0.1.2 · epoch 2026-09-23 / C69

Audit stamp: **dirty** · unaudited

Scaffold for a future course — not ready to teach

Canonical source: `textbook/helut-living-textbook.tex`. Corpus of record:  
`directives/claim-sheet.md`. This volume is an AI-bootstrapped collecting place for theorems and  
formulae. Humans must vet, bulk, fill, and edit it before it is a textbook. Frontier chapters are labeled  
frontier.

## Draft in preparation — not yet ready to teach from

Edition 0.1.2, epoch 2026-09-23 / C69.

This volume is being *prepared* as a textbook, and is written in the voice of the book it is intended to become. It therefore may contain text that implies a course already exists — chapters that lecture, exercises that are set, labs that are specified, a syllabus that assigns weeks and grading weights. Please read that as work in progress. The material has not been taught, has not been reviewed by anyone qualified to sign off on the pedagogy, and is not part of any curriculum.

Status is stated here once and stamped on each page rather than repeated inside every paragraph, so that the text can be read — and grown — as continuous prose.

**What is nonetheless load-bearing.** Editorial readiness and factual correctness are separate. Every **C**, **H** and **N** reference, every **Reproduced** box, and every theorem in this book is bound to `directives/claim-sheet.md` and a reproduce receipt, and that binding is checked mechanically by `Scripts/claim_audit.py` in CI. Claims, lemmas, theorems and formulae are not invented to fill the scaffold: a human checks each against the sheet before commit. If a sentence in this book disagrees with the claim sheet, **the sheet wins** and the sentence is a bug.

Numbers carry the same discipline. Timings are medians of repeated runs with the spread printed where they have been re-measured (`Scripts/bench_repeat.py`); single-observation figures inherited from earlier work are being replaced incrementally.

# How to read the living boxes

Seven boxes appear throughout this book. They are not decoration; they are the difference between a textbook and a press release. Editorial status is not their job — that is stated once on the previous page and stamped on every page footer. These boxes carry *evidential* status, which is a separate axis and the one that must never slip.

**Reproduced (Cn).** A result that someone else can re-run from `REPRODUCE.md` and the claim inventory. The prose may treat it as established *in this corpus*. Keyed to a **C** row, always — never to free text and never to an **H** id alone.

**Open hedge (Hn).** An honest asterisk. Say it with the hedge attached. Do not let it get dropped in a problem set or a paper.

**Measured negative (no C row).** A measured *failure*: something we ran that did not pass. Printing clean negatives is doctrine, so these appear on purpose — but they are never green, and they are never receipts.

**Re-checked, second model (second model).** Independently re-derived by a different model reading the same evidence. Weaker than a human check and stronger than nothing. Never a substitute for **Reproduced**.

**Frontier — not a claim.** Speculative territory from `directives/potential-avenues.md`. A seminar essay, never a midterm fact.

**This section grows when.** A stub waiting on a new **C** row, a closed **H**, or a graduated avenue. Skip, or read the matching directive.

**Not a course yet.** Section-level reminder that a chapter is skeleton rather than finished text. The book-level statement of that is the draft notice on the previous page; this box marks the specific places where the scaffolding shows.

**Why theorems carry two numbers.** The corpus — `directives/`, the claim sheet, the papers — has called these results *Theorem 1*, *Theorem 1''*, *Theorem 2* and *Theorem 3* since before this book existed, and external notes cite them that way. L<sup>A</sup>T<sub>E</sub>X numbers theorems by chapter. Rather than break either convention, each theorem prints both: Theorem 9.1 of this book is also corpus

Theorem 1. If you are following a reference from the claim sheet, match on the corpus name; if you are reading straight through, use the chapter number.

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# Part I

## How This Book Lives

# Chapter 1

## Preface

This book is a *scaffold* for a subject that did not have a name until the laboratory forced one. It is not a finished textbook. It is not ready to teach.

Universities already teach *reconfigurable computing*: LUTs, interconnect, place-and-route, FPGA fabrics, high-level synthesis. They already teach *applied cryptography*: AES, RSA, lattices, sometimes a lecture on fully homomorphic encryption. They almost never teach what happens when the reconfigurable fabric *is* a machine-learning graph API, the configuration is a Yosys netlist (not an FPGA bitstream—see Chapter 5), and the values on the wires are torus samples that must remain encrypted while a host clock ticks flip-flops.

That intersection is the intended subject of a future course. We call the working stack **HELUT**—Homomorphic Edge Look-Up Tensors—and we call the subject **reconfigurable homomorphic computing**.

The stack has three pillars [7, 3]:

1. **Netlist-clocked torus FHE**. Gate-level sequential circuits evaluate as exact  $\mathbb{Z}/2^{32}\mathbb{Z}$  tensor graphs, with real bootstrapping-key blind rotation and machine-checkable certificates.
2. **Differentiable hardware cryptanalysis**. Discrete INIT tables melt into continuous tensors; a  $\lambda$ -squeeze grades shatter versus hold; reciprocal structure that combinatoric sieves miss can reappear as an involution.
3. **Adversarial polymorphic ciphers**. Blue ciphers co-evolve under Red pressure and fail closed, rather than waiting to become the next historically leaked stream cipher.

**Why a living textbook.** Research papers freeze a slice of a laboratory. This course cannot. The claim inventory [2] moves: Metal kernels get faster, hedges close, avenues stay unlabeled until they earn receipts. If this scaffold ever becomes a course, a professor who taught from a PDF dated June would be teaching a different Metal compiler than a professor teaching from August. So the book is versioned against the corpus *epoch* (2026-09-23 / C69 in this edition) and is honest about stubs.

**What this edition is.** Edition 0.1.2 is an AI-bootstrapped collecting place: theorems, formulae, claim IDs, and lab receipts as they land—aspirational and future-forward, not a syllabus you can run next semester. Pillar chapters exist so humans have something to vet, bulk up, fill in, and rewrite. It is a long-term project. Where the laboratory is thin, the page says so.

**What this edition is not.** It is not a claim that HELUT invented TFHE [1, 10]. It is not a production key-size manual. Sage filled the estimator JSON (**C23**); **H1** remains because four of eight calibration anchors disagree with Cost **rop** by more than 16 bits, which is why “176-bit secure” is not a sentence this corpus supports. It is not a decrypt of U-534 / P1030680. It is not a side-channel paper. Those sentences are printed again in Chapter 3 because students will forget them the week before the final.

**This edition is a draft, and it is AI-generated.** The TeX tree was bootstrapped with AI under human architectural direction (see `AI_DISCLOSURE.md`). That is not the same as human authorship of a course. Chapters will be wrong—sometimes in large ways—until people diff them against the claim sheet, fill the stubs, and rewrite the lecture voice. If a box and the sheet disagree, the sheet wins; file an issue or patch the `.tex`. Do not treat an epoch bump as proof the PDF was ready for a classroom. **Do not hallucinate the math.** A new theorem, lemma, or claim ID is a draft until a human has checked it against the inventory and a log; it does not belong in a git commit until then—at least going forward. We will have to catch up on commits up to now.

**Who wrote it.** The HELUT project at Digital Defiance. The scientific voice of the book is the same voice as the research-release doctrine: lead with what runs, and with what it does not prove. The prose itself still needs human editors.

**How to cite an epoch.** Cite the git commit and the claim-sheet epoch, not “the HELUT textbook” as if it were a static ISBN. When a student reproduces **C18**, they are reproducing a dated Metal NTT persist tile, not an eternal constant of nature.

# Chapter 2

## To the instructor

Do not teach from this PDF. This chapter is a briefing for a course that does not exist yet: it is not in the ACM/IEEE curriculum guidelines, and this book is not ready to put in front of students. The pages that follow are an aspirational skeleton—grading weights, tracks, a fifteen-week outline—so that when humans have vetted and filled the rest of the volume, the pedagogy has a place to live. Until then, treat this chapter as a design note, not an instructor’s manual.

### 2.1 The course in one paragraph

Students learn to treat a *synthesized gate netlist* as the universal intermediate representation of reconfigurable computing, to lower that netlist onto a *tensor engine* (here: Apple Silicon **MPSGraph**), and to evaluate it either as a boolean-faithful oracle or as torus FHE with certificates. They then learn a second compiler direction—melting discrete hardware into continuous tensors—and a third: designing ciphers that are born under adversarial pressure. The moral of the course is not “FHE is fast on a laptop.” The moral is: *reconfigurable computing has a new fabric, and honesty about what that fabric proves is part of the engineering.*

### 2.2 Prerequisites

---

|                     |                                                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Required</b>     | Digital logic (LUTs, flip-flops, timing) [12]; linear algebra; a first crypto course (modular arithmetic, attack models) [14].       |
| <b>Helpful</b>      | One HDL (Verilog or VHDL); one systems language; Yosys as a black box [17].                                                          |
| <b>Not required</b> | Prior FHE, prior Metal, prior Swift, prior lattice cryptography. The torus chapter is self-contained at the level this course needs. |

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### 2.3 Two tracks

Not every department can put an M-series Mac in front of every student.

**Systems track.** Apple Silicon laboratory. Students build, tick, and certificate-read HELUT from `REPRODUCE.md`. Labs 13 assume this track.

**Theory track.** No GPU required. Students work the five-cell test, noise lemmas, and published logs as *data*. A department can still grade a claim-audit final (below).

A mixed offering works: theory problem sets for everyone; systems labs as a project option or a second section.

## 2.4 Suggested grading

| Weight | Instrument                         | Why it exists                                                                                                                    |
|--------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 30%    | Problem sets                       | Negacyclic arithmetic, packing, certificate reading.                                                                             |
| 40%    | Laboratories or theory equivalents | Receipts, not vibes.                                                                                                             |
| 20%    | <b>Claim audit</b>                 | Take a public sentence; run the five-cell test from Chapter 3. Fail the assignment if the student upgrades a hedge to a theorem. |
| 10%    | Frontier essay                     | Pick one avenue from Chapter 15. Must label it frontier. Must propose the <i>first experiment</i> that would graduate it.        |

The claim audit is the signature assignment of this course. It trains the skill the field currently lacks: refusing to launder an open hedge into a keynote.

## 2.5 What to lecture, what to assign as reading

- **Lecture** Parts II–III (foundations and Pillar I), the honesty chapter, and Theorem 10.1 (**C19**), which is stated in Chapter 10 — this chapter only refers to it.
- **Seminar** Pillar II empirics (shatter / hold, campaign as case study) and Pillar III in edition 0.1.2. Do not lecture shatter grades as if they were Theorem 10.1.
- **Do not lecture as fact** Chapter 15. Give it as the last two seminars, with the frontier boxes left on the slides.

## 2.6 How to pull a new edition mid-semester

Science will move under your feet. That is the point of a living textbook.

1. Diff `directives/claim-sheet.md` against the epoch printed on the title page.
2. If a new **C** row appeared, assign the matching lab receipt as a one-page addendum; do not pretend the old PDF contained it.
3. If an **H** closed, tell the class explicitly: “this asterisk is gone; here is the log.”
4. If only frontier text moved, do not change the midterm.

Bump `\livingepoch` in `textbook/preamble.tex` when you accept an addendum. Cite git.

## 2.7 Classroom norms

- Trivial / mock torus encodings are a *shape laboratory*, not FHE. Students who call the boolean oracle “encrypted computing” lose points on purpose.
- **Campaign cryptanalysis** (Enigma / P1030680) is a *case study* in Pillar II methods, not the FHE claim, and not a decrypt.
- **Do not lecture** HELUT as an unsupervised or mobile agent, a self-rewriting FPGA, or a production encrypted CPU. Those sentences fail the five-cell test of Chapter 3.
- **H4** Grade B is covering Track A (**C52–C54**), not native  $k=1$ , not **cryptoPublicMS**, not PicoRV covering at production  $N$ .
- Speculative systems (encrypted LLM firewalls, self-modifying dark binaries, honey-token ledgers) are ethics-and-design discussions, not implementation homework, until they have **C** rows.

**This section grows when.** A published instructor’s guide with slide skeletons, a Moodle/Canvas quiz bank, and a non-Apple reference implementation of the negacyclic oracle for the theory track.

# Chapter 3

## The living contract

### 3.1 Why a textbook needs a contract

Ordinary textbooks accrete errata. This one accretes *science*. Without a contract, the lecture slides of 2027 will still be quoting a fused **MPSGraph** megagraph that the laboratory killed in August 2026, or—worse—will quote a speculative avenue as if it were a theorem.

The contract is the research-release doctrine [7] translated into *eventual* pedagogy. Until humans have edited this volume into a real textbook, the boxes below are how the scaffold refuses to launder hedges—not a license to open a class.

### 3.2 The three inventories

Every public sentence in HELUT is classified in `directives/claim-sheet.md`:

| Tag       | Name                | Classroom meaning                                                                                            |
|-----------|---------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Cn</b> | Reproducible result | You may lecture it. Students may cite it. Someone else can re-run it [2].                                    |
| <b>Hn</b> | Open hedge          | You may discuss it. You must keep the asterisk. A problem-set “true/false” that drops the asterisk is false. |
| <b>Nn</b> | Non-implication     | You must actively prevent this inference. It is a common exam trap.                                          |

A fourth class lives only in this book and in `directives/potential-avenues.md`:

|          |                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------|
| Frontier | Not a claim until it has a <b>C</b> row and a reproduce command.<br>Seminar fuel. Never a midterm fact. |
|----------|---------------------------------------------------------------------------------------------------------|

### 3.3 The five-cell test

For an atomic claim  $C$ , the doctrine requires all five, or  $C$  is hedged:

1. **Proof** — lemma, reduction, or graded empirical protocol with pass/fail.
2. **Table** — parameters, comparisons, or grades.

3. **Metric** — a number with units and a bar.
4. **Examples** — at least two concrete, runnable instances.
5. **Application** — at least one end-to-end reason to care.

**Example 3.1** (Five cells for **C18**). Metal 3-prime NTT persist blind-rotate. Proof: CPU+Metal NTT  $\equiv$  schoolbook (19 tests) plus microbench PASS bits. Table:  $N \in \{64, 1024\}$  s/BR versus schoolbook persist and fused-EP. Metric: 0.433 s/BR at  $N = 1024$  (gpu 0.43 s, RSS 148 MiB). Examples: microbench identity LUT; full\_adder boolean SING (**Stimuli In, Golden out**—encrypted evaluation must match the cleartext netlist on the same generic vectors; see Chapter 7). Application: production-shaped polynomial degree on a laptop GPU graph.

**Example 3.2** (Five cells for **C19**). TensorLUT continuous $\rightarrow$ discrete, corpus Theorem 1, stated as Theorem 10.1 in Chapter 10 [9]. Proof: six `TensorLUTFormal.check*` lemmas, aggregated by `certificate()`. Table: lemma vs `holds`; campaign grades kept separate. Metric:  $F_{\text{crypto}} = 0$  on unmutated baseline;  $\pi = 0$  on binary INIT. Examples: M4 baseline emit; freeze-core involution; blind 3-pair PASS. Application: a stecker search that cannot propose a non-reciprocal map. Asterisk: structural, not a U-534 break (**H6**).

**Exercise 3.1** (Claim audit). Take any sentence from a blog post, a grant abstract, or last year’s slide deck about “encrypted computing on a Neural Engine.” Fill the five cells or mark the sentence a non-claim. If you cannot name the log file, it is not a **C** row.

### 3.4 Disclosure versus discovery

|              | Disclosure package                                 | Trajectory                   |
|--------------|----------------------------------------------------|------------------------------|
| Goal         | Someone else re-runs what we assert                | We know what to invent next  |
| Artifact     | <code>REPRODUCE.md</code> , logs, reproduced boxes | trajectory file, Chapter 14  |
| Failure mode | Unreproducible hype                                | Silence / no next experiment |

When a trajectory item graduates, it earns a **C** row (or a tightened **H**) and a reproduce command—*then* it may enter lecture voice. That sentence is the entire editorial policy of this book. An AI draft of a lemma is not a graduation: a human must check the statement before it is committed.

### 3.5 Standing non-claims

Print these on the course website.

**Non-claim.** HELUT does not introduce a new lattice assumption, does not claim to have invented TFHE, and does not offer “176-bit secure” as a slide title. **C23** filled the Sage JSON (production HELUT 175.7 vs Sage 180.2 on `prod-n1024-s16`); **H1** is the remaining  $|\Delta| > 16$  anchors and the core-SVP vs Cost `rop` split. Trivial/oracle Metal graphs are not FHE. Covering Track A Grade B (**C52–C54**) is not native  $k=1$  and not `cryptoPublicMS`. `publicMS` gadgets are not a side-channel theorem. Campaign fitness is not encrypted tick rate. TensorLUT involution grades are not a U-534 break. Quantum attacks are out of scope. Metal/GPU power analysis is parked [6].

## 3.6 How a chapter grows

1. A laboratory result lands in a log under `logs/`.
2. A **C** or **H** row is updated in the claim sheet.
3. The matching chapter gains an example, a table row, or a converted box; `\livingepoch` bumps.
4. The syllabus appendix does *not* automatically change. Instructors opt in (Chapter 2).

**This section grows when.** Automated extraction of claim-sheet tables into Appendix B, so the snapshot cannot drift. Until then, humans diff the sheet against the appendix before each teaching epoch.

## 3.7 A verdict that varies is a bug, never a parameter

On 2026-08-15 a re-validation sweep ran every claim in this corpus five times instead of once. Two rows stopped agreeing with themselves. One of them, **C69**, was recorded as a clean negative — an encrypted adder that failed to evaluate correctly at LWE dimension  $n=512$  — and it reproduced only twice in five attempts.

The tempting explanation was physical, and this book had all the vocabulary for it: the measured residual sat at 82% of the decode half-gap  $\delta/2$ , so noise was presumably straddling the threshold. That story was written down. It was wrong.

The disproof was already in the tool’s own output. Across runs of the same command the measured residual was *byte-identical* —  $B_{\text{bk}} = 858\,907$  every time — and every seed in the harness is a hardcoded constant. Identical inputs were producing different outputs.

*Honesty note 3.1.* Before reaching for a physical explanation of an inconsistent result, check whether the computation is deterministic. A fixed-seed pipeline that returns two answers has a software defect, and no amount of noise analysis will find it.

The defect was mundane and worth knowing about in any language with hash-based dictionaries. Encrypting the primary inputs walked a `Dictionary` while drawing from a shared serial random-number generator:

```
for (port, bits) in inputs {                                // hash order
    wires[wire] = encryptLWE(..., rng: &rng) // shared RNG
}
```

Swift randomises dictionary hash seeds per process, so every run assigned a different mask vector to a different wire. Ciphertexts differed, the noise realisation differed, and wherever the margin was thin the decode became a coin toss. Sorting the iteration fixed it, and the withdrawn negative became a pass.

Three details generalise beyond this repository:

- **The concurrency was innocent.** The obvious suspect was a parallel evaluation loop added for throughput. Forcing it serial changed nothing; every shared cache on the path held a correct lock. The fault was in ordinary single-threaded code.

- **Only a marginal experiment could see it.** Under a noiseless bootstrap key the blind rotation cancels the input mask exactly, so the outputs are identical whichever mask was drawn. The bug was invisible except where noise made masks matter — which is to say, invisible except near a bar.
- **One run cannot detect it.** Hash-order dependence is a property of a *process*, so it does not appear in a single receipt, in code review, or in a test suite that runs each case once.

**Non-claim.** This section is not a claim that the corpus is now deterministic everywhere. It is a claim about one fixed defect, one withdrawn negative, and a method: run it more than once, and treat disagreement as a defect until proven otherwise.

**This section grows when.** A determinism assertion covers the encrypted path end to end — for example, byte-identical ciphertexts across processes under a fixed seed, checked in CI rather than by hand.

# Part II

## Foundations of the Subject

# Chapter 4

## A new subject

### 4.1 The gap between two mature fields

Reconfigurable computing classically means: a designer writes RTL, a synthesizer emits LUTs and flip-flops, a bitstream configures an FPGA, and the fabric evaluates the circuit at a clock [12, 11]. The fabric is spatially wired. The configuration is a bitstream. The values on the wires are bits.

Fully homomorphic encryption classically means: a scheme evaluates circuits *on ciphertexts* [1, 10]. The “fabric” is an arithmetic library—NTT butterflies, key-switching decompositions, bootstrapping. The configuration is a parameter set. The values on the wires are polynomials in  $\mathbb{Z}_q[X]/(X^N + 1)$ .

Each field has textbooks. The composition does not.

**Definition 4.1** (Reconfigurable homomorphic computing). The study of compiling *already-synthesized* sequential netlists onto *tensor engines* so that LUT evaluation, flip-flop clocking, and (when claimed) torus FHE occupy the same graph object, with explicit certificates of what the graph does and does not prove.

HELUT is one laboratory realization of that definition: Yosys JSON in, MPSGraph out, host posedge clock, LWE/GLWE samples on the FHE path [3]. Apple Silicon was the *first* engine because the architect needed that graph hardware to brute-force the  $\lambda$ -squeeze and synthesis loops — not because the subject is a Mac API. The definition does not name Apple. A CPU-only restatement of Theorem 1 is `python3 Scripts/tensorlut_math_ref.py`. CUDA and other fabrics remain trajectory until they have C rows (`directives/why-apple-silicon.md`).

### 4.2 Three questions that open the course

1. **Representation.** Can a commodity ML graph API host exact modular arithmetic at  $q = 2^{32}$ , or does the API silently round?
2. **Clocking.** Can sequential logic—enables, sync resets, a RISC-V core—tick on that graph without the host becoming a bit-banged FPGA?
3. **Honesty.** When the same object grows a real bootstrapping key, what certificates must ship with every tick, and which sentences does the evidence still not support?

Pillar I is the constructive answer to (1)–(3). Pillar II asks a fourth question the FPGA curriculum never asked: *what if the INIT table is not binary?* Pillar III asks a fifth: *what if the circuit under evaluation is allowed to mutate because the attacker in the next room is the same compiler?*

### 4.3 Why “reconfigurable” is the right word

An FPGA is reconfigurable because the LUT INIT bits and the interconnect are late-bound. HELUT is reconfigurable in the same sense, with three substitutions:

| Classical FPGA     | This course                          | Consequence                          |
|--------------------|--------------------------------------|--------------------------------------|
| LUT INIT bitstream | Yosys <code>\$lut</code> truth table | Same IR                              |
| Routing fabric     | Tensor edges in one graph            | Place-and-route becomes lowering     |
| Clock tree         | Host <code>graph.run</code> loop     | Timing closure becomes tick protocol |
| IOB / BRAM         | Placeholders and MTLBuffers          | Memory is unified, not dual-port     |
| Encrypted?         | Optional; certificate-gated          | Oracle $\neq$ FHE                    |

The last row is the pedagogical load-bearing wall. Students who have built FPGAs want to say “the CPU is encrypted” the moment PicoRV32 ticks. The boolean oracle path *does* clock PicoRV32 on ciphertext-*shaped* vectors (**C1**). The FHE path *does* fetch NOPs on encrypted PicoRV32 at demo  $N=8$  (**C47** CPU, **C51** Metal): the PC walks, decrypted  $Q$  matches cleartext, and calibrated hardness is 4.0 bits—not the production 175.7 row. Covering noisy BK at  $N=1024$  on that core is not claimed. Chapter 9 exists so that confusion is a grading event rather than a press cycle.

### 4.4 The three pillars as a curriculum, not a brand

Roadmap language names them the Turing pillar, the Schneier pillar, and a grand challenge on proving circuits [8]. In this course they are simply I, II, III, in the order a student can verify them:

- Without I, II is a float32 netlist toy and III has nowhere to run.
- Without II, I is a compiler course with FHE homework.
- Without III, I and II only attack the past; they do not design the next cipher under fire.

Edition 0.1.2 treats Pillar I in lecture depth, Pillar II Theorem 10.1 (**C19**, stated in Chapter 10) in lecture depth with empirics in seminar, Pillar III in seminar depth, and the grand-challenge / side-channel items as labeled frontier (Chapter 15).

### 4.5 Related systems, scoped not ranked

| System                | Exact $q=2^{32}$ Metal | Netlist clock | BK <code>\$lut</code> FHE | Diff. hardware |
|-----------------------|------------------------|---------------|---------------------------|----------------|
| Concrete / tfhe-rs    | —                      | compiler IR   | yes                       | —              |
| OpenFHE / HELib       | —                      | circuit API   | yes                       | —              |
| HELUT boolean oracle  | yes                    | yes           | —                         | —              |
| HELUT encrypted path  | yes                    | yes           | yes                       | —              |
| TensorLUT (Pillar II) | —                      | Yosys INIT    | —                         | yes            |

This is a *scope* matrix, not a throughput bake-off [3]. The encrypted-path “yes” is GGSW `$lut` FHE as certified (**C4–C6**, covering **C52–C54**); it is not PicoRV covering at production  $N$ . Concrete is a production TFHE compiler; HELUT is a netlist-clocked tensor fabric with an FHE path. Students comparing “ms/gate” across those rows without naming encoding,  $N$ , and certificate status fail the five-cell test.

**Exercise 4.1.** Write four sentences, one per row of the matrix, that a journalist could print without creating a **N**. Then write the one-sentence lie each row most tempts.

**This section grows when.** A fourth pillar chapter, if FHE-gate / ZK-depth work in the mid-term trajectory [8] graduates. Until then it is Chapter 14, not a `\part`.

## Chapter 5

# Netlists as the universal IR

### 5.1 From RTL to LUTs

A combinational Boolean function  $f : \{0,1\}^k \rightarrow \{0,1\}$  is exactly a truth table of  $2^k$  bits. FPGA vendors package that table as a  $k$ -input look-up table. Yosys [17] emits the same object after `abc -lut 2` (or 4, or 6) as a cell of type `$lut` with a parameter string `INIT`.

**Definition 5.1** (Yosys `$lut`). A combinational cell whose output bit is `INIT` indexed by the concatenated input bits. HELUT treats every `$lut` as one programmable gate, whether the backend is a multilinear oracle or a TFHE blind rotate.

Sequential logic is a `$_DFF*` (or enable / sync-reset variant). The clock is not a net in the tensor graph: it is a *host protocol* (Chapter 7). That substitution—clock as protocol, not as a routed tree—is the first reconfigurable-computing heresy of the course, and it is load-bearing. Tensor engines do not give you a global posedge. They give you `graph.run`.

### 5.2 Why we start from synthesized netlists

High-level FHE compilers start from a language and lower to homomorphic ops. HELUT starts from a *netlist that already exists*: PicoRV32, a regex DFA, a decision tree, an Enigma core [16]. The pedagogical reason is the same as the systems reason:

- The IR is stable. Verilog front-ends change; `$lut` does not.
- Sequential depth is explicit. Multiplicative depth in an arithmetic circuit is a metaphor; DFF count is a fact.
- Any HDL, any synthesizer that can emit Yosys JSON, is in scope.
- Cryptanalysis targets (historical machines, stream ciphers) already *are* netlists.

*Remark 5.1.* Starting from netlists does not make HELUT “better than Concrete.” It makes it a different compiler problem: *spatial* Boolean fabric onto a graph API, not *scalar* program onto PBS.

### 5.3 The four objects a student must be able to draw

1. A **LUT** as a box with `INIT` and  $k$  wires.

2. **A DFF** as a box whose  $Q$  at tick  $t + 1$  is a mux of  $D$ , enable, and reset, evaluated *between* graph runs.
3. **A batch axis  $B$**  as  $B$  independent copies of the same fabric in one run—the reconfigurable analog of SIMD, and the reason a Bombe search is a tensor problem.
4. **A polynomial degree  $N$**  as the *ciphertext shape*, not the circuit size.  $N = 1$  is clear-shape boolean.  $N = 1024$  is TFHE-shaped. Confusing  $N$  with LUT count is the most common homework error in week 2.

## 5.4 Worked lowering: a half-adder

A half-adder is two gates:  $\text{sum} = A \oplus B$ ,  $\text{carry} = A \wedge B$ . Under *additive torus* / *mock* encoding, XOR is a vector add (free). AND is a LUT (or a PBS). On the encrypted GGSW path, both gates are blind-rotates. Do not lecture “XOR is free in FHE” from this paragraph.

**Example 5.1** (Half-adder as a graph). Yosys emits (typically) one `$lut` for the AND and uses an add/xor cell or a second LUT for the sum, depending on techmap. HELUT’s Phase-2 compiler [3] lowers this to placeholders for  $A, B$ , an add node, a LUT node, and output binds—one persistent `MPSGraph`. There is no FPGA routing step. The “place-and-route” *is* the tensor edge list.

**Exercise 5.1.** Synthesize a 1-bit full adder with Yosys (`abc -lut 2`) and count `$lut` versus `$_DFF*` cells. Then argue, in four sentences, why the encrypted SING of that netlist is a better first FHE lab than PicoRV32. (PicoRV32 is **C1** on the *oracle* path and **C45–C51** on encrypted demo  $N=8$  at 4.0 bits; the encrypted path’s production- $N$  multi-LUT workhorse is the adder, **C6** and **C20**; covering noisy adder at the same  $N$  is **C52**.)

## 5.5 Batch is a first-class capacity knob

On an FPGA, duplicating a circuit costs LUTs. On a tensor engine, duplicating a circuit is a batch dimension  $B$  on every placeholder. Memory scales as the broadcast intermediates, not as “more CLBs.”

**Reproduced (C1).** Yosys `$lut/DFF`  $\rightarrow$  Metal/CPU tensor eval on the cleartext / trivial-oracle path. PicoRV32, Enigma stream, and campaign-scale batch search all sit on this lowering.

At  $N = 1024$ ,  $B = 1000$ , Enigma M3’s TFHE-shaped working set is gigabytes while clear-shape  $N = 1$  stays flat. Students designing a “just crank  $B$ ” attack are designing a memory bomb. That is a reconfigurable-computing lesson, not an FHE lesson.

**Exercise 5.2.** Using only the boolean-path scale table in the HELUT README (epoch of this edition), plot tick time versus  $B$  at  $N = 1$  and  $N = 1024$ . Identify the regime where graph overhead dominates  $N$ , and the regime where  $N = 1024$  becomes memory-bound.

## 5.6 What this chapter refuses to call a bitstream

A Metal graph executable is late-bound, cached, and device-specific. It is tempting to call it a bitstream. We do not, in this edition. A bitstream has a documented configuration memory; an `MPSGraphExecutable` has an Apple runtime. The analogy is useful in lecture and false on an exam. Chapter 8 returns to this when the compiler stops unrolling schoolbook arithmetic into MLIR and starts calling ring kernels.

## Chapter 6

# Torus arithmetic and programmable bootstrapping

This chapter is a working minimum of TFHE [1, 10], not a substitute for a lattice-crypto course. The goal is that a student can read a HELUT certificate line without bluffing.

### 6.1 The torus and the native word

**Definition 6.1** (Discrete torus used here). HELUT freezes the modulus  $q = 2^{32}$  and represents torus elements as `UInt32`. Addition is native wraparound. This is a *systems* choice: the machine word *is* the torus.

That freeze answers a common question:  $q = 2^{32}$  is *not* a proof that lattice FHE requires a 32-bit modulus, and TensorLUT (**C19**) is already Boolean /  $[0, 1]$ . Binary-modulus FHE is a new experiment, not a silent reread of **C4–C6**. Split: `directives/q-32-vs-q-2.md`.

The ring of polynomials is

$$R_q = \mathbb{Z}_q[X]/(X^N + 1),$$

the negacyclic ring. Multiplication by  $X$  is a rotation with a sign flip on wrap-around—the algebraic fact that makes blind rotation a barrel shifter on coefficients.

*Honesty note 6.1.* Exact mod  $2^{32}$  integer tensors are not “homomorphic encryption.” They are the arithmetic. Encryption begins when those tensors are LWE or GLWE *samples* with a secret and noise.

### 6.2 LWE samples, in one page

**Definition 6.2** (LWE sample [15]). For secret  $s \in \mathbb{Z}_q^n$ , a sample is  $(a, \langle a, s \rangle + e + m) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$ , with  $a$  uniform and  $e$  small Gaussian (or discrete) noise. GLWE is the polynomial analog: mask polynomials plus a body polynomial in  $R_q$ .

HELUT’s encrypted path uses LWE/GLWE samples and GGSW bootstrap keys. Decision-LWE is bound to IND-CPA by a machine-checkable certificate (**C5**); the bit estimate at production  $(n, \sigma) = (1024, 2^{16})$  is HELUT calibrated 175.7 versus Sage Cost 180.2 on that named row (**C23**) and is *not* “176-bit secure”—print **H1** with the number.

## 6.3 Programmable bootstrapping, sketched

A TFHE Boolean gate is often evaluated by programmable bootstrapping (PBS): an encrypted phase selects a coefficient of a *test polynomial* via *blind rotation*, then a sample extract returns an LWE encryption of the LUT output.

Blind rotation is a product of CMUXes, one per bit of the encrypted phase. Each CMUX is an external product against a GGSW encrypt of a secret bit. The accumulator lives in  $R_q$  (a GLWE). The sequential dependence across the  $N$  (or  $n$ ) bits is real: you cannot parallelize away the chain inside one bootstrap. You *can* refuse to unroll each external product into  $O(N)$  host MLIR nodes. That refusal is Chapter 8.

## 6.4 Negacyclic Toeplitz embedding

Let  $A(X) = \sum_{i=0}^{N-1} a_i X^i$ . Negacyclic convolution against a vector  $x$  is  $M_A x$ , where  $M_A$  is the Toeplitz matrix whose first column is  $(a_0, \dots, a_{N-1})^\top$  and each subsequent column is the previous column rotated downward with the wrapped entry *negated* [13].

**Example 6.1** ( $N = 4$  by hand). Let  $A = (a_0, a_1, a_2, a_3)$ . Then

$$M_A = \begin{pmatrix} a_0 & -a_3 & -a_2 & -a_1 \\ a_1 & a_0 & -a_3 & -a_2 \\ a_2 & a_1 & a_0 & -a_3 \\ a_3 & a_2 & a_1 & a_0 \end{pmatrix}.$$

On `UInt32`,  $-a \equiv 2^{32} - a$ . The matrix product *is* the ring product. HELUT’s Phase-1 kernel proved this bit-exact against a CPU schoolbook oracle, including torus corners  $\{0, 1, 2^{31}, 2^{32} - 1\}$ .

**Exercise 6.1.** Expand  $M_A$  for  $A = (1, 2, 3, 4)$  over  $\mathbb{Z}/2^{32}\mathbb{Z}$  and multiply by  $x = (1, 0, 0, 0)^\top$ . Interpret the output as a polynomial. Then multiply by  $x = (0, 1, 0, 0)^\top$  and explain the sign that appears.

**Why dense Toeplitz is not the FHE hot path.** Materializing  $N \times N$  at  $N = 1024$  is  $\Theta(N^2)$  words per LUT—a kernel proof and a memory stress, not a production PBS. The encrypted hot path moved through tiled schoolbook, fused external product, GPU-resident tiles, and 3-prime NTT (**C16–C18**). Students implementing “the HELUT paper kernel” as their FHE engine are implementing 2026-era Phase 1, which the compiler chapter treats as a *control-plane ancestor*.

## 6.5 Message spacing and $Z_{2N}$

Blind rotation addresses coefficients at granularity  $\delta = q/(2N)$ . Packing several LWE bits into one phase must respect the cyclic group  $Z_{2N}$ , not an integer that merely looks like a power.

**Reproduced (C6).** Multi-LUT encrypted full\_adder SING PASSes at  $N \in \{256, 512, 1024\}$ ; this is what closed **H2** on 2026-08-12. The bug was arity-3 packing overflowing  $256 \cdot 2N$  headroom; the fix is rotation-native encrypt plus `packLWEbits` reduce mod  $2N$ , with `rotationPower` preferring  $Z_{2N}$  representatives.

That closed hedge is the best homework in the chapter: a systems bug that is also an algebraic bug.

**Exercise 6.2.** Suppose  $N = 256$  so  $2N = 512$ . A pack of three bits each scaled by 256 can leave the  $2N$  range. Show a numeric counterexample, then write the reduction that restores a valid rotation power.

## 6.6 Noise, in the sense this course needs

Three layers ship as code, not as vibes (**C5**):

|                                     |                                                                                                                                                                                                                                                                                                          |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TFHENoiseProof                      | Discrete $\infty$ -norm inject lemmas.                                                                                                                                                                                                                                                                   |
| Gaussian $\varepsilon$ -certificate | Ingest failure $\leq 2^{-64}$ under independent-noise hypotheses.                                                                                                                                                                                                                                        |
| TFHENoisyBKCertificate              | Depth under a BK-noise bound $B_{\text{bk}}$ . Default noiseless Track A SING still uses $e = 0$ BK. Covering Track A at $N=1024$ , torus $\sigma=128$ , $k=7$ is <b>C52–C54</b> ( <b>H4</b> Grade B). <code>cryptoPublicMS</code> inject remains <b>C26/C56</b> ; native $k=1$ remains <b>C37/C55</b> . |

*Honesty note 6.2.* A noiseless BK is a correct *functional* bootstrap and an incomplete *depth* story if that is all you ran. **C22** is the covering-gadget measurement at  $N \leq 128$ . **C52–C54** put covering noisy BK on  $N=1024$  Metal SING at  $\sigma=128$ ,  $k=7$ . Students who say “we have production noisy BK” from the  $N = 8$  table, or who drop the native- $k=1$  / `cryptoPublicMS` remainders, are failing **H4**.

## 6.7 Parameters you may quote

From the cookbook [4], production-shaped boolean:

| Knob                 | Value                                                                                                      | Asterisk           |
|----------------------|------------------------------------------------------------------------------------------------------------|--------------------|
| $N$                  | 1024                                                                                                       |                    |
| $q$                  | $2^{32}$                                                                                                   | native wor         |
| $\sigma$             | $2^{16}$                                                                                                   | $\ll \delta/2 = 2$ |
| Target $\varepsilon$ | $\leq 2^{-64}$                                                                                             | Gaussian u         |
| Classical target     | $\geq 128$ bits                                                                                            | HELUT 17           |
| $B_{\text{bk}}$      | covering Track A <b>C52–C54</b> ; native $k=1$ <b>C37/C55</b> ; <code>cryptoPublicMS</code> <b>C26/C56</b> | <b>H4</b> remain   |
| Inter-LUT refresh    | <code>publicMS</code> default                                                                              | lattice-com        |

Demo  $N = 8$  will not meet 128-bit hardness. The certificate reports that honestly. Using demo  $N$  for a “secure encrypted CPU” slide is a **N**.

## Part III

# Pillar I — Netlist-Clocked Torus FHE

# Chapter 7

## Netlist-clocked torus FHE

Pillar I is the reason the course exists: a synthesized sequential circuit becomes one tensor graph, ticked by the host, with an encrypted backend that is actually LWE.

### 7.1 The HELUT question

*Can Apple Silicon’s **MPSGraph** host exact torus modular arithmetic and gate-level sequential circuits—and can that same hardware object carry real bootstrapping keys without lying about it?*

The first half is a systems existence proof. The second half is a certificate problem. HELUT answers both, with the oracle path and the FHE path named as different objects [3, 7].

### 7.2 Pipeline

$$\begin{array}{c} \text{Verilog} \xrightarrow{\text{Yosys}} \text{JSON netlist} \xrightarrow{\text{HELUTCore}} \text{one MPSGraph} \\ \text{Host Swift loop} = \text{posedge clock; DFF } Q \text{ ping-pongs across ticks} \\ \$\text{lut backend} = \text{oracle (multilinear / trivial PBS)} \quad \text{or} \quad \text{encrypted (BK blind rotate)} \end{array}$$

All FHE-path arithmetic tensors remain `MPSDataType.uInt32`. `Float32` belongs to `TensorLUT` (Pillar II), not to this chapter.

### 7.3 Compiler lowering

`YosysGraphCompiler` walks a `write_json` module:

1. **Inputs.** Each input port bit becomes an `InputNode` placeholder of shape  $[B, N]$ .
2. **State.** Each recognized flip-flop allocates a  $Q$  placeholder *before* LUT lowering so sequential feedback nets resolve.
3. **LUTs.** Cells schedule until all drivers exist; combinational cycles abort.
4. **Next-state.** Enables and sync-resets become exact  $\mathbb{Z}/2^{32}\mathbb{Z}$  muxes, e.g. active-high enable

$$Q_{\text{next}} = E \cdot D + (1 - E) \cdot Q.$$

5. **Outputs.** Driven nets bind to port tensors; Yosys "x" becomes constant 0.

Cell coverage includes `$_DFF*`, `$_DFFE*`, `$_SDFF*`, `$_SDFFE*`, `$_SDFFCE*`—the techmap PicoRV32 actually emits.

## 7.4 Host clock

Each tick is one `graph.run`. With `retainHistory=false`, two preallocated state buffer sets ping-pong and output scratch is reused, so a 1000-tick stress test does not leak `MTLBuffer` retention. Primary port feeds (including scripted `resetn`) are host-mutable shared buffers; LUT matrices or BK packs are uploaded once and cached.

This is the reconfigurable clock tree: not routed silicon, but a protocol with explicit buffer ownership.

## 7.5 SING as the teaching metric

**SING** expands to **Stimuli In, Golden out**.

In hardware verification, a *golden model* is the flawless reference circuit: feed test vectors (stimuli) in, expect the exact right bits out. For HELUT the golden model is the cleartext Yosys netlist simulator. A SING check (`-sing` / `-bench-encrypted-metrics`) is the primary teaching metric and the laboratory analogue of a scan chain: encrypt the same generic stimuli, evaluate under GGSW blind-rotate, decrypt, and demand bit-exact agreement with cleartext.

A passing encrypted tick under SING is not only a boolean match. The log is graded as a cryptographic bundle:

- **Functional:** ciphertext decrypts to the clear netlist (the hard gate).
- **Discrete:**  $\infty$ -norm inject lemmas hold where the path prints them.
- **Gaussian:** ingest failure probabilities stay under stated hypotheses.
- **Hardness:** calibrated classical bit estimates print with **H1** attached until the estimator row is honest.
- **Depth:** noisy bootstrap-key residuals are modeled or measured; default Track A throughput SING may still use  $e=0$  BK (**H4**); covering noisy SING receipts are separate claims (**C28**, **C33–C37**, **C41**, **C43**, **C52–C54**).

Brutal engineering rule: if SING fails, you do not get to discuss milliseconds. Throughput and timing benches are irrelevant until the FHE path certifies its own correctness against the golden netlist.

**Laboratory 7.1** (Encrypted adder SING). Run the cookbook SING at demo  $N$  and at production-shaped  $N$  (Appendix D). Record pass/fail, ms/row, and which certificate types printed. Then answer: which of those numbers would you be willing to put in an abstract without an asterisk?

## 7.6 Two backends, one netlist

| Backend                             | Claim status                     | Use in this course                                  |
|-------------------------------------|----------------------------------|-----------------------------------------------------|
| Trivial / multilinear / CMUX oracle | Shape laboratory, <i>not</i> FHE | Batch scaling, PicoRV32 boot, Enigma letter clocks. |
| encrypted (LWE + GGSW BK)           | Pillar I FHE claim (C4–C6)       | SING vs cleartext; certificates on every tick.      |

Inter-LUT refresh defaults to `publicMS` (lattice-compatible BK masks). `.secret` remains for coverage. `.none` is a debug foot-gun.

**Reproduced (C4–C6).** Encrypted LWE/GLWE + GGSW blind-rotate per `$lut`; encrypted  $\equiv$  clear multi-netlist SING; `full_adder` through  $N = 1024$  (`publicMS` and `secret`).

**Reproduced (C38).** Encrypted host-clocked DFF  $Q$  on a 4-bit counter ( $6 \$lut + 4 \$\_DFFE\_PP\_$ ): CPU  $N=8$  and Metal  $N=1024$  `public-ms` boolean SING PASS. Not PicoRV32; fused metal-netlist and `cpu-ggsw` remain combinational-only.

**Reproduced (C53).** Same counter under covering-b1 noisy BK,  $N=1024$ , torus  $\sigma=128$ ,  $k=7$  stride encoding (C52): Metal secret 110.35s/2 and `public-ms` 67.98s/2 both PASS. Receipt: `logs/helut-encrypted-n1024-metal-sing-counter-covering-b1-gauss-sigma128-k7-stride.log`.

**Reproduced (C40).** Encrypted toy ISA: 4-bit accumulator, NOP or ADD `imm`, 11 `$lut + 4 DFFs`. Metal  $N=1024$  SING PASS.

**Reproduced (C54).** Same toy ISA under covering-b1 noisy BK,  $N=1024$ ,  $\sigma=128$ ,  $k=7$ : Metal secret 194.10s/2 and `public-ms` 104.18s/2 both PASS. Receipt: `logs/helut-encrypted-n1024-metal-sing-toy-isa-covering-b1-gauss-sigma128-k7-stride.log`.

**Reproduced (C45).** Encrypted PicoRV32: 4785 `$lut + 1565 DFFs`. CPU `public-ms` boolean output SING PASS at demo  $N=8$ , one host posedge, 76.12ms/row, hardness 4.0 bits.

**Reproduced (C46).** Encrypted PicoRV32 10-tick `resetn` boot (hold 3). PASS 0.972s/10 (97.20ms/row) with  $Q \equiv$  clear. Plain `$_DFF_P_` must copy native  $D$  (no second `public-MS`).  $N=32$  1-tick also PASS (819ms/row, 16-bit hardness);  $N=64$  1-tick PASS (4.35s/row, 32-bit hardness).

**Reproduced (C47).** Encrypted PicoRV32 NOP ROM: always-ready `addi x0,x0,0`. Ten instruction fetches, `mem_addr 0,4,...,0x24`,  $Q \equiv$  clear. 3.15s/32 at demo  $N=8$ . Receipt: `logs/helut-encrypted-n8-cpu-sing-picorv32-nop-fetch.log`.

**Reproduced (C49).** Encrypted PicoRV32 tiny program: `addi x1,x0,1, NOP, sw x1,0(x0)`. Store `wdata=1` into host RAM. PASS 3.79 s/48 (78.90 ms/row) at demo  $N=8$ . `$_SDFFCE_*` enable gates reset; `rdata` held for NBA `mem_xfer`; host DFF copies native LWE.

**Reproduced (C50).** Same ROM plus `lw x2,0(x0)`. Host RAM on the NBA `mem_xfer` cycle: `rdata=1`. PASS 3.77 s/48 (78.53 ms/row). First-cycle LOAD log may still show leftover NOP 0x13; PicoRV samples the xfer. Receipt: `logs/helut-encrypted-n8-cpu-sing-picorv32-prog-lw.log`.

**Reproduced (C51).** Metal PicoRV32 NOP-fetch at demo  $N=8$ : tiled-kernel BR (`-metal-br-tile 8`; fused default traps). Two fetches (0, 4). PASS 63.55 s/8 (7.94 s/row). Per-LUT Metal BR, not fused metal-netlist.

## 7.7 Why PicoRV32 is real in FHE, with a speed qualifier

PicoRV32 lowers to 4785 LUTs and 1565 DFFs [16]. On the boolean-oracle path it compiles in  $\sim 1.3$  s and ticks at  $\sim 173$  ms ( $N = 1024$ ,  $B = 1$ ) in the 2026-08-12 boolean bench (**C1**). That is a fabric existence proof: a commodity neural/GPU graph can host a soft CPU's *shape*.

The encrypted path uses the same host-clocked sequential seam as the toy counter (**C38**) and toy ISA (**C40**). Claim **C45** is one encrypted host posedge of the full PicoRV32 netlist at demo  $N=8$  (output SING). Claim **C46** is the PRD 10-tick scripted `resetn` boot at the same  $N$ : CPU public-ms boolean SING PASS in 0.972 s (97.20 ms/row), calibrated hardness 4.0 bits, with decrypted  $Q$  matching the clear register file. A second public-MS on already-native  $D$  for plain `$_DFF_P_` cells had been silently folding  $1 \rightarrow 0$  — outputs could still match. Claim **C49** is execute+store: `addi x1,x0,1` then `sw, wdata=1`, 3.79 s/48 (78.90 ms/row). Claim **C50** is load-back: `lw` sees stored 1 on the NBA xfer, 3.77 s/48 (78.53 ms/row). Claim **C51** is the first Metal sequential PicoRV receipt: NOP-fetch, two instruction addresses, tiled BR, 7.94 s/row at demo  $N$ . Metal  $N=1024$  is not SING-logged this epoch — the LUT-tax is the remaining engineering, not a missing compiler. *Honesty note 7.1.* “We executed encrypted RISC-V” means **C50** on CPU (addi/store/load of 1) and **C51** on Metal only as a NOP-fetch at demo  $N$  ( $\sim 8$  s/row). Not Linux; not Metal  $N=1024$ . Avenue 2 in Chapter 15 is self-modifying dark code, not this receipt.

## 7.8 Threat / fidelity split

|                                   |                                                                                                                                                                                                                                                                    |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| In scope for the <i>datapath</i>  | Exact modular eval; shape $[B, N]$ ; DFF routing; Yosys coverage for PicoRV32.                                                                                                                                                                                     |
| In scope for the <i>FHE claim</i> | LWE/GLWE + BK; noise / $\varepsilon$ / hardness / noisy-BK <i>certificates</i> as implemented.                                                                                                                                                                     |
| Out of scope unless measured      | Side channels; quantum; estimator Cost as “secure bits” on every row ( <b>H1</b> ); native $k=1$ / <code>cryptoPublicMS</code> noisy BK and PicoRV covering at $N=1024$ ( <b>H4</b> remainder; Grade B is <b>C52–C54</b> ; PicoRV covering-b2 is <b>C60</b> FAIL). |

The split is intentional. If the tensor datapath cannot boot a CPU under mock encodings, adding real TFHE noise only makes the systems problem harder. If the FHE path cannot SING an adder, a CPU slide is theatre.

**Reproduced (C58).** PicoRV32 abc -lut 6 netlist: 2006 LUT + 1565 DFF. CPU public-ms boolean 1-tick SING PASS at  $N=64$  (1.35 s, 32-bit). Demo  $N=8$  cannot host LUT6 tables.

**Reproduced (C59).** Same lut6 netlist after sequential wavefront: CPU boolean 0.165 s/1 at  $N=64$  ( $\sim 8.2\times$  vs **C58**).

**Reproduced (C60).** Covering-b2  $k=7$  Metal  $N=1024$  1-tick Q SING FAIL after  $\sim 33$  min of combinational BRs.

**Reproduced (C62).** Metal PicoRV lut6  $N=1024$   $e=0$  public-ms boolean 1-tick Q SING PASS (373.89 s). Covering remains **C60/C61**.

**Reproduced (C63).** Covering-b2 noisy BK on that lut6 core PASSes Q at  $N=64$  (1.72 s). The  $N=1024$  covering Q fail is torus-scale depth, not a lut6/wavefront race.

**Reproduced (C64).** Extract $\rightarrow$ KS closes public-MS PBS at  $n=64$ . Native- $k$  covering-b2 adder and counter SING at poly  $N=1024$ . PicoRV covering remains **C60**.

**Reproduced (C65).** Same lut6 PicoRV, covering noisy BK, Metal  $N=1024$ , extract $\rightarrow$ KS  $n=64$ : Q SING PASS (b2 114 s, b1 212 s). Walk-around of **C60/C61**, not a silent close of  $n=N$   $k=7$ .

**Reproduced (C66).** 10-tick covering boot on that core: PASS 1136 s/10.

**Reproduced (C67).** KS  $n=128$  covering adder PASS;  $n\geq 256$  SIGTRAP.

**Reproduced (C68).** NOP-fetch 8 ticks: PASS 911 s/8, PC  $0x0\rightarrow 0x4$ .

**Reproduced (C69).** Covering KS adder SING PASSes at  $n=256$  and  $n=512$ . The  $n=512$  case was recorded as a FAIL until 2026-08-15; that negative is **withdrawn**. It was an artifact of nondeterministic input encryption, not a noise limit — see §3.7.

**This section grows when.** PicoRV 1w; KS  $n=512$  with  $k>1$ ;  $n=N$  native  $k$ ; cryptoPublicMS.

## Chapter 8

# The Metal torus compiler

### 8.1 The picture that changed the compiler

```
Yosys $lut --> pack LWE --> blind rotate (N CMUXes)
|
+-- failure: each external product expands
|   sum_j b_j * X^j a as O(N) MPSGraph ops
|   --> millions of host MLIR nodes
|   --> encode wall >> run wall
|
+-- design: small graph that *calls* ring kernels
      encode cheap; run on Metal; cache
```

CMUX dependence across bootstrap bits is sequential and real. The pain was never “CMUX cannot run on a GPU.” The pain was *encoding a schoolbook expansion that should never have been a graph* [5].

**Measured negative (fused  $N = 1024$  DNF — no C row).** Schoolbook-in-MPSGraph at  $N = 1024$  spent 11.6 h on host encode, never reached the GPU, and died **SIGTERM 143**. A compiler that does not finish is a measurement, not an anecdote — but it is a *negative*, it has no claim-sheet row, and it must never be read as a reproduced result.

### 8.2 Doctrine

SoftBus / ANE remains a graph machine. We stop mistaking “unroll all ring math into MLIR” for “use the GPU.” MPSGraph is a *staged IR* and a *scheduler*. Polynomial multiplication and the external product are first-class GPU operations.

Why this GPU first: the laboratory was an M-series Mac pushed to extremes so the  $\lambda$ -squeeze could run at all ([directives/why-apple-silicon.md](#)). That does not freeze the math to Apple.

### 8.3 Phase 1 — survive $N = 1024$

Phase 1 does not change the cryptography. It changes the control plane.

| Lever              | Idea                                                                  | Teaching point                                                 |
|--------------------|-----------------------------------------------------------------------|----------------------------------------------------------------|
| Tiled blind rotate | CMUX windows of width $W$ ; re-ingest ACC                             | Encode cost scales with $W$ , not fused $N \times$ schoolbook. |
| Constant CSE       | Shared splat banks per tile                                           | Graph builders die on <code>constantWithScalar</code> spam.    |
| Executable cache   | Compile once per $(N, \ell, W, BK)$                                   | Trial 1 is cold; ticks $\geq 2$ must hit cache.                |
| Telemetry          | <code>encode</code> / <code>gpu_run</code> / <code>host_repack</code> | Silence is not a hang if the log counts tiles.                 |

Phase 1 without Phase 2 still leaves  $O(W \cdot N)$  encode inside each tile. Phase 2 without Phase 1 is a kernel drop into an unmeasured fused path. The laboratory did both, in that order.

## 8.4 Phase 2 — ring kernels

1. **Negacyclic poly-mul** as a Metal kernel, bit-identical to CPU schoolbook (**C18** NTT  $\equiv$  schoolbook, 19 tests).
2. **Fused external product (C16)**: one `helut_ggsw_external_product` launch per CMUX.  $N = 1024$  microbench 1.043 s/BR.
3. **GPU-resident BR tile (C17)**: ACC + BK on device; 0.519 s/BR at  $N = 1024$ ;  $N = 64$  at 0.001 s/BR.
4. **3-prime NTT persist tile (C18)**: gadget digits in-tile; BK in NTT domain; pointwise EP + iNTT + CRT. 0.433 s/BR at  $N = 1024$  (gpu 0.43 s, RSS 148 MiB).

Default Metal BR: fused if  $N \leq 64$ , tiled-kernel otherwise (NTT EP inside tiles). Legacy fused megagraph is `-metal-br-fused` only, and at production  $N$  it is a museum exhibit.

## 8.5 Numbers a student may quote (epoch 2026-09-23 / C69)

| Path at $N = 1024$                     | s/BR (micro)          | SING / 8 rows        | RSS          |
|----------------------------------------|-----------------------|----------------------|--------------|
| Fused schoolbook megagraph             | DNF (11.6 h)          | —                    | $\sim 4$ GiB |
| Poly-mul kernel                        | 3.645                 | —                    | —            |
| Fused EP ( <b>C16</b> )                | 1.043                 | 25.1 s (3.14 s/row)  | —            |
| Persist schoolbook ( <b>C17</b> )      | 0.519                 | 12.2 s (1.52 s/row)  | 68 MiB       |
| NTT persist ( <b>C18</b> ) serial SING | <b>0.433</b>          | 15.6 s (1.95 s/row)  | 148 MiB      |
| Wavefront-parallel ( <b>C20</b> )      | 0.420 (fused 3-prime) | 10.6 s (1.33 s/row)  | —            |
| Crypto $\ell = 2$ SING ( <b>C21</b> )  | —                     | 11.38 s (1.42 s/row) | —            |
| Covering-b1 $k=7$ adder ( <b>C52</b> ) | —                     | 29.40 s/2 public-ms  | —            |
| Covering-b2 $k=7$ adder ( <b>C57</b> ) | —                     | 10.33 s/1 public-ms  | —            |
| CPU SING full_adder ( $e=0$ BK)        | —                     | $\sim 52$ s          | —            |

**Reproduced (C20, C21).** Wavefront-parallel independent \$1ut BRs: boolean SING 10.6 s/8 beats persist-schoolbook **C17**; noiseless cryptoPublicMS  $\ell = 2$  SING 11.38 s/8 vs **C14**'s 175.6 s. **C21** is  $e=0$  BK. Noisy cryptoPublicMS is **C26/C56**.

**Open hedge (H3).** The scheduler, not a faster NTT, is what beat **C17** on SING. Serial in-tile NTT (**C18**) still loses boolean SING (15.6 s). Sage **H1** is independent of these wall-clocks. Do not quote 0.43 s/BR as a PicoRV covering tick: covering-b1  $\ell=32$  is a different gadget tax (**C52**).

**Exercise 8.1** (Compiler post-mortem). In 400 words: why did the fused megagraph never reach the GPU? Your answer must distinguish encode wall from run wall, and must not claim that “Metal cannot multiply polynomials.”

**Exercise 8.2** (Reading **H3**). Is **C18** a success or a failure? Defend a one-sentence answer that a claim-audit grader will accept. (Hint: both, along different bars.)

## 8.6 Punchline, for the lecture slide

Phase 1 makes encoding *finish*. Phase 2 makes encoding *irrelevant*. Together they are a graph-native torus compiler: staged IR + ring kernels + cached executables—instead of a schoolbook megagraph hoping the GPU will eventually appear. (SoftBus is Pillar III’s cipher contract, not this Metal lowering.)

**This section grows when.** NTT EP inside the crypto  $\ell = 2$  tile (C21’s SING log still shows  $W = 64$  schoolbook tiles); then **H3** is only Sage **H1**.

## Chapter 9

# Certificates as first-class artifacts

Most compiler courses grade a binary that runs. This course grades a binary that *refuses* when it cannot certify.

### 9.1 The surface

| Certificate                                     | Role                                                                                                         |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| TFHENoiseProof                                  | Discrete $\infty$ -norm inject lemmas.                                                                       |
| TFHEAsymptoticSecurityCertificate               | Gaussian ingest $\varepsilon \leq 2^{-64}$ .                                                                 |
| TFHELWEHardnessCertificate                      | Decision-LWE $\rightarrow$ IND-CPA + bit estimate.                                                           |
| TFHELWECalibration                              | Anchor table for the estimator.                                                                              |
| TFHENoisyBKCertificate / TFHENoisyBKMeasurement | Depth under $B_{\text{bk}}$ ; identity residual $\hat{\sigma}$ ( <b>C22</b> ; covering Grade B <b>C52</b> ). |
| TFHELWEEstimatorProtocol                        | External lattice-estimator merge.                                                                            |

`TFHENoise.refuse` if no certificate. That is not an API flourish. It is the difference between a demo and a claim.

### 9.2 How to read a SING log

**SING** = **S**timuli **I**n, **G**olden out (Section 7.5): encrypted evaluation must match the cleartext netlist on the same generic vectors. A passing encrypted tick is then read as a bundle:

- Functional: ciphertext decrypts to the clear netlist (the SING gate).
- Discrete:  $\infty$ -norm lemmas held on the inject.
- Gaussian: ingest  $\varepsilon$  under the stated hypotheses.
- Hardness: calibrated classical bits, with **H1** attached (**C23** filled the estimator JSON; four anchors still  $|\Delta| > 16$ ).
- Depth: noisy-BK model; **C22** measured at covering gadget; **C26** graded fail at `cryptoPublicMS`  $N=1024$ ; covering Track A torus-scale is **C52–C54**; native  $k=1$  remains **C37/C55**; `cryptoPublicMS` noisy SING remains **C26/C56**.

A lab report on this material is incomplete if it stops at milliseconds: the certificate lines are the result, and the timing is context for it.

**Reproduced (C22).** Identity-LUT residual under a covering gadget fills  $B_{\text{bk}}$  and  $\hat{\sigma}$ . At  $N = 8$ , inject  $B = 64$  gives  $\hat{\sigma} \approx 6396 \ll \delta/2$ . At  $N = 128$  the residual is still  $\infty$ -norm decodable but the union  $\varepsilon$  is nowhere near  $2^{-64}$ . Re-measured 2026-08-15 across sample sizes:  $\hat{\sigma} \rightarrow 1.67 \times 10^6$  and union  $\varepsilon \log_2 \rightarrow -18.0$  with a 95% upper bound of  $-16.1$  at  $n = 512$ . The archived  $-23.5$  came from  $n = 4$  and was optimistic small-sample noise (§9.6).

**Reproduced (C26).** Product-shaped  $N=1024$  identity residual: inject  $B=64$  undecodable on `cryptoPublicMS/crypto`; inject  $B=4$  on `.crypto` may be  $\infty$ -norm OK with  $\varepsilon \log_2 \approx -1$  (not  $-64$ ). **C56:** stride- $k=7$  does not turn this into Metal SING. Receipt: `logs/helut-noisy-bk-measure-n1024.log`.

**Reproduced (C28).** Track B: Metal `cryptoPublicMS full_adder SING` at covering  $N=128$  with inject  $B=64$  PASSes ( $B_{\text{bk}} \approx 1.62 \times 10^6$  decodable). Receipt: `logs/helut-encrypted-n128-metal-sing-crypto-noisy.log`.

**Reproduced (C27).** Theorem 3: under  $q = 2^{32}$ , exact public-MS covering forces  $(1 + \log_2 N) \mid 32$ ; among practical degrees  $\{8, \dots, 2048\}$  only  $N \in \{8, 128\}$  qualify.  $N=1024$  `cryptoPublicMS` has  $\text{baseLog} \cdot \ell = 11 \cdot 2 = 22 \neq 32$ . Statement: `directives/ggsw-public-ms-covering.md`. Reproduce: `swift test -c release -filter testGGSWPublicMSCoveringCertificate`.

**Theorem 9.1** (Exact public-MS covering degrees; corpus Theorem 3). *Let the torus word be  $q = 2^{32}$  and let a public modulus-switch gadget have  $g_0 = \delta$ . If the gadget covers, then  $(1 + \log_2 N)$  divides 32. Consequently, among  $N \in \{8, 16, \dots, 2048\}$  the only exact covering degrees are  $N = 8$  and  $N = 128$ .*

**Proof (machine-checked).** `GGSWPublicMSCovering.certificate()` asserts the divisibility lemma and enumerates `exactDegrees = [8, 128]` over the practical range; the test additionally pins  $\text{baseLog} \cdot \ell = 22$  for  $N=1024$ .

**Reproduced (C29).** Theorem 3': for any power-of-two torus word  $w$ , exact public-MS covering among practical degrees stays  $\{8, 128\}$ . Widening the limb does not unlock  $N=1024$  ( $\text{baseLog} = 11$  never divides  $2^k$ ). Same receipt as **C27**; lemma `powerOfTwoWordObstruction`.

**Reproduced (C30).** At covering  $N=128$ , inject  $B \leq 8$  prints  $\varepsilon \log_2 = -\infty$ , but  $\hat{\sigma}$  remains  $\sim 10^4$ – $10^5$ . At  $B=64$ ,  $\varepsilon \log_2 \approx -24$  (**C22**). Receipt: `logs/helut-noisy-bk-eps-sweep-n128.log`.

**Reproduced (C31).** Incomplete public-MS gap:  $N=1024$  `uncoveredBits = 10`. Approx candidate is covering `.crypto` ( $\text{baseLog} = 8$ ,  $g_0 \neq \delta$ ). Statement: `directives/ggsw-incomplete-covering.md`.

**Reproduced (C32).** At  $N=1024$ , `.crypto` inject  $B=1$  is  $\infty$ -norm decodable;  $\varepsilon \log_2 \approx -8.4$  at 8 trials (not  $-64$ ); `cryptoPublicMS` remains undecodable for  $B \geq 1$ . Receipts: `logs/helut-noisy-bk-measure-n1024-B1.log`, `logs/helut-noisy-bk-measure-n1024-B1-t8.log`.

**Reproduced (C33).** Track A Metal SING with covering `.crypto` at  $N=1024$ , inject  $B=1$ : secret-refresh and public-ms covering paths both PASS (non-zero decodable  $B_{bk}$ ). Receipts: `logs/helut-encrypted-n1024-metal-sing-covering-crypto-noisy-B1.log`, `logs/helut-encrypted-n1024-metal-sing-covering-publicms-noisy-B1.log`.

**Reproduced (C34).** Finer covering `baseLog= 4` at  $N=1024$ : inject  $B=1$  yields  $\hat{\sigma} \approx 2.95 \times 10^4$ , asymptotic  $\varepsilon \log_2 \approx -913$  (meets  $\leq 2^{-64}$ ); Metal secret + public-ms covering-b4 both PASS. Receipts: `logs/helut-noisy-bk-covering-b4-n1024-B1.log`, `logs/helut-encrypted-n1024-metal-sing-covering-b4-noisy-B1.log`. Torus-scale  $B=128$  on the same gadget is still undecodable.

**Reproduced (C35).** Covering `baseLog= 2` at  $N=1024$ :  $\varepsilon \leq 2^{-64}$  holds through inject  $B=16$  ( $\varepsilon \log_2 \approx -65.4$ ;  $B=32$  drops to  $\approx -40.5$ ). Metal secret + public-ms covering-b2 at  $B=16$  both PASS ( $B_{bk} \approx 1.86 \times 10^5 / 6.15 \times 10^4$ ). Receipts: `logs/helut-noisy-bk-covering-b2-eps-vs-B.log`, `logs/helut-encrypted-n1024-metal-sing-covering-b2-noisy-B16.log`. Torus-scale  $B \sim 128$  remains  $\varepsilon \approx -1.6$ .

**Reproduced (C36).** Covering `baseLog= 1` at  $N=1024$ :  $\varepsilon \leq 2^{-64}$  holds through inject  $B=32$  ( $\varepsilon \log_2 \approx -139$  at 4 trials;  $B=64 \approx -26$ ). Metal secret + public-ms covering-b1 at  $B=32$  both PASS. Receipts: `logs/helut-noisy-bk-covering-b1-n1024-B32.log`, `logs/helut-encrypted-n1024-metal-sing-covering-b1-noisy-B32.log`. Finest covering does *not* unlock  $B=128$  ( $\varepsilon \approx -0.6$ ).

**Reproduced (C52).** Stride- $k$  encoding: public-MS is native- $\delta$  (wires in  $\{0, k\}$ ), test poly  $T[k \cdot \text{addr}]$ . Covering-b1 at  $N=1024$ , torus  $\sigma=128$ ,  $k=7$ : 8-trial  $\varepsilon \log_2 \approx -170$ ; Metal secret 44.08s/2 and public-ms 29.40s/2 both PASS. Receipts: `logs/helut-noisy-bk-covering-b1-gauss-sigma128-n1024-k7-stride-t8.log`, `logs/helut-encrypted-n1024-metal-sing-covering-b1-gauss-sigma128-k7-stride.log`. Native  $k=1$  remains **C37**; `cryptoPublicMS` remains **C26**.

**Reproduced (C55).** Native  $k=1$  covering-b1  $\sigma=128$  at polynomial  $N=1024$ : cutting LWE  $n$  (CMUX count) makes the identity residual decodable at  $n \leq 512$ , but  $\varepsilon \log_2$  is  $-36.6$  ( $n=64$ ),  $-10.9$  ( $n=256$ ),  $-6.5$  ( $n=512$ ) — none  $\leq -64$ . Default  $n=N=1024$  stays undecodable (**C37**, 8 trials). Not a hardness-preserving close of native spacing.

**Reproduced (C56).** `cryptoPublicMS` plus stride- $k=7$  at inject  $B=1$ : identity residual becomes decodable ( $\varepsilon \log_2 \approx -12.6$ ), but Metal public-ms full\_adder SING fails (sum mismatch). Torus  $\sigma=128$  remains undecodable. Do not quote  $k\delta$  as a `cryptoPublicMS` noisy-BK close.

**Reproduced (C57).** Covering-b2  $\ell=16$ ,  $k=7$ ,  $\sigma=128$  at  $N=1024$ : identity  $\varepsilon \log_2 \approx -110.7$  (4 trials). Metal public-ms adder 10.33 s/1 and regex 23 LUT 26.69 s/1 PASS. Covering-b4 public-ms adder and E256 58-LUT covering-b2 SING FAIL. CPU covering-b2 PASSes at demo  $N=8$ .

**Reproduced (C58).** PicoRV32 abc -lut 6: 2006 \$lut ( $-58\%$  vs lut2 4785). CPU public-ms boolean SING PASS at  $N=64$  (1.35 s, 32-bit).  $N=8$  SIGTRAPs because table size  $64 > N$ . Not covering; not production  $N$ .

**Reproduced (C59).** Sequential combinational wavefront: independent ready \$luts concurrentPerform even with DFFs. PicoRV lut6  $N=64$  CPU boolean 0.165 s/1 versus **C58** 1.35 s ( $\sim 8.2\times$ ).

**Reproduced (C60).** PicoRV lut6 covering-b2  $k=7$  Metal  $N=1024$  one tick: combinational BRs finished in  $\sim 33$  min; Q SING FAIL (\$\_DFF\_P\_ want= 0 got= 1). Timed negative.

**Reproduced (C61).** Same PicoRV lut6 at covering-b1  $k=7$ :  $\sim 61$  min then Q SING FAIL on a nearby \$\_DFF\_P\_. Finer gadget did not fix sequential Q.

**Reproduced (C62).** Metal PicoRV lut6  $N=1024$  public-ms boolean  $e=0$  one tick: Q SING PASS in 373.89 s. Hardness 175.7 with **H1**. Not covering.

**Reproduced (C63).** Same lut6 netlist, covering-b2  $\sigma=128$ , CPU  $N=64$  one tick: Q SING PASS in 1.72 s ( $B_{bk} = 88782$ ). Native  $k$ . Stride  $k=7$  at this  $N$  SIGTRAPs. Does not close **C60**.

**Reproduced (C64).** Extract $\rightarrow$ KS ( $n=64 \ll kN$ ):  $\delta$ -lattice GLev. CPU covering-b2  $\sigma=128$  native  $k=1$  at poly  $N=1024$ : adder 4.22 s, counter 6.35 s. Not **LWE-176**. **C37** still at  $n=N$ .

**Reproduced (C65).** PicoRV lut6 covering Metal  $N=1024$  extract $\rightarrow$ KS  $n=64$  one tick: Q SING PASS covering-b2 114 s, covering-b1 212 s. **C60/C61** remain  $n=N$   $k=7$  negatives.

**Reproduced (C66).** Same flags, 10-tick resetn boot: Metal PASS 1135.8 s/10 (113.6 s/row). Q SING. Idle mem.

**Reproduced (C67).** Covering-b2 KS adder ladder at poly  $N=1024$ :  $n=128$  PASS 8.58 s;  $n=256$  and  $n=512$  SIGTRAP after extract $\rightarrow$ KS.

**Reproduced (C68).** NOP-fetch 8 ticks on lut6 PicoRV covering KS  $n=64$ : Metal PASS 911 s/8, fetches 0x0, 0x4.

**Reproduced (C69).** Identity trials= 1 at **LWE**  $n \geq 256$ : covering-b2 adder PASS at  $n=256$  (17.1 s) and at  $n=512$ . C67 SIGTRAP was four identity BRs. The  $n=512$  sum mismatch recorded here until 2026-08-15 was a determinism artifact and is **withdrawn** (§3.7).

### 9.3 Calibration is not estimation

| Label          | $n$  | $\sigma$ | HELUT | Estimator    | $ \Delta $ |
|----------------|------|----------|-------|--------------|------------|
| demo-N8        | 8    | $2^{12}$ | 4.0   | 33.1         | 29.1       |
| weak-n256      | 256  | $2^{17}$ | 40.4  | 53.7         | 13.3       |
| mid-n512       | 512  | $2^{16}$ | 95.5  | 92.4         | 3.2        |
| classic-n630   | 630  | $2^{15}$ | 129.0 | 106.5        | 22.5       |
| n768-s16       | 768  | $2^{16}$ | 135.6 | 135.4        | 0.2        |
| prod-n1024-s16 | 1024 | $2^{16}$ | 175.7 | <b>180.2</b> | 4.5        |
| n1024-s17      | 1024 | $2^{17}$ | 160.7 | 189.9        | 29.3       |
| n2048-s16      | 2048 | $2^{16}$ | 336.1 | 369.8        | 33.7       |

$q = 2^{32}$ , binary secret, native Sage 10.9 + lattice-estimator (**C23**). Merge tolerance is 16 bits: four rows pass, four fail.

**Open hedge (H1).** Quote “ $\sim 176$  calibrated classical bits at production  $(n, \sigma)$ ” together with “estimator 180.2 on prod-n1024-s16.” A midterm bubble that says “176-bit secure” without naming the estimator row, or that treats every calibration anchor as estimator cost, is wrong.

### 9.4 Five-cell test, applied to certificates

Certificates are themselves claims. They need the five cells:

1. Proof: the lemma in code (TFHENoiseProof, ...).
2. Table: the calibration table; the cookbook production row.
3. Metric:  $\varepsilon \leq 2^{-64}$ ; bit estimates;  $B_{\text{bk}}$ .
4. Examples: demo  $N = 8$  (honest fail of 128-bit) and production  $N = 1024$ .
5. Application: refuse-closed encrypt so a netlist tick cannot silently drop hardness.

**Exercise 9.1. C52** closed covering Track A noisy BK at  $N=1024$  ( $\sigma=128$ ,  $k=7$ ). Name the two remainders of **H4** that are still graded negatives, and the two follow-up rows that mapped them without closing  $\varepsilon$  or SING (**C37/C55**, **C26/C56**). PicoRV covering at production  $N$  is the third remainder; it is a LUT-tax, not a missing SING log at demo  $N$ .

### 9.5 Honesty as an engineering interface

The non-claims of Chapter 3 are not ethics theater. They are API documentation for speech:

- Oracle Metal graphs expose the *shape* of encrypted evaluation (batch,  $N$ , clock). They do not expose Decision-LWE.
- publicMS is an on-lattice refresh intent. It is not a leakage proof (**N** on side channels).
- A campaign’s settings/second is a cleartext Metal number. It does not certify encrypted tick rate.

**Laboratory 9.1** (Certificate literacy). Run `.build/release/helut -hardness-table` and classify each printed row as demo, weak, or production-shaped. For each row, write one sentence a journalist may print and one sentence they may not.

**This section grows when.** The four off-tolerance anchors (demo-N8, classic-n630, n1024-s17, n2048-s16) get a documented estimator model, or HELUT’s core-SVP fit is revised; then **H1** can shrink further.

## 9.6 Why $\varepsilon$ needs a sample size

The failure probability  $\varepsilon$  that appears on a noise certificate is not sampled. It is an analytic Gaussian tail evaluated at a *measured* standard deviation:

$$\varepsilon = \Pr[|Z| \geq \delta/2], \quad Z \sim \mathcal{N}(0, \hat{\sigma}^2), \quad \log_2 \varepsilon \approx -\frac{(\delta/2)^2}{2\hat{\sigma}^2 \ln 2}.$$

Read the exponent carefully:  $\log_2 \varepsilon \propto -1/\hat{\sigma}^2$ . Differentiating, a relative error  $r$  in  $\hat{\sigma}$  moves  $\log_2 \varepsilon$  by about  $2|\log_2 \varepsilon|r$ . At  $|\log_2 \varepsilon| = 65$  a one percent error in  $\hat{\sigma}$  is worth 1.3 orders.

And  $\hat{\sigma}$  is an RMS over however many residuals were collected — one per trial. The standard error of a standard-deviation estimate from  $m$  samples is about  $1/\sqrt{2m}$ , so:

| $m$   | $\text{SE}(\hat{\sigma})/\hat{\sigma}$ | unresolved orders at $\log_2 \varepsilon = -65$ |
|-------|----------------------------------------|-------------------------------------------------|
| 2     | 50.0%                                  | $\pm 65$                                        |
| 4     | 35.4%                                  | $\pm 46$                                        |
| 8     | 25.0%                                  | $\pm 32$                                        |
| 64    | 8.8%                                   | $\pm 11.5$                                      |
| 1 024 | 2.2%                                   | $\pm 2.9$                                       |
| 8 450 | 0.77%                                  | $\pm 1$                                         |

*Honesty note 9.1.* A measured  $\varepsilon$  of  $-65.4$  against a  $2^{-64}$  bar, taken at four samples, carries roughly  $\pm 46$  orders. It does not clear the bar. It does not fail to clear the bar. It says nothing about the bar at all. Quote the confidence bound, or quote nothing.

The bound is standard:  $m\hat{\sigma}^2/\sigma^2 \sim \chi^2(m)$ , so a one-sided upper bound on  $\sigma$  is  $\hat{\sigma}\sqrt{m/\chi_\alpha^2(m)}$ , and  $\varepsilon$  evaluated there is the conservative figure — the only one worth comparing against a target. HELUT prints it:

```
cryptoPublicMS B=64: ... eps log2=-23.5
[n=4, 95%up=-2.8 DOES NOT CLEAR -64, +/-1 needs n~1403]
```

**Re-checked, second model (C22 at  $N=128$ ).** Re-derived on 2026-08-15 by a second model reading the same evidence, then re-measured on the original hardware. This is weaker than a human check and stronger than a single archived number; the distinction is the point of this box.

**Reproduced (C22).** Worked example. At covering  $N=128$  with inject  $B=64$ ,  $\hat{\sigma}$  converges to  $1.67 \times 10^6$  and union  $\varepsilon \log_2$  to  $-18.0$  (95% bound  $-16.1$ ) by  $n = 512$ , at 0.21 s per trial. The conclusion — nowhere near  $2^{-64}$  — was right all along; it is now resolved rather than asserted.

*Honesty note 9.2* (Sample cost can make a bar unreachable). The same measurement at production degree costs about 155 s per trial ( $N=1024$ , covering-b1,  $\ell=32$ ). The  $n \approx 8450$  needed for  $\pm 1$  order is therefore roughly *fifteen days*. Some bars cannot be established by running the experiment longer, and saying so is part of the certificate.

**This section grows when.** The residual measurement samples all  $N$  coefficients of the GLWE accumulator instead of only the extracted one. Each is a draw from the same noise distribution, so that is about a thousandfold more samples per bootstrap at no extra cryptographic cost, and it would turn the fifteen days above into minutes.

**Exercise 9.2.** A colleague reports  $\varepsilon \leq 2^{-64}$  from three trials and proposes shipping the parameters. Compute the unresolved orders. Then compute how many trials would be needed to support the sentence they wrote, and how long that would take at 155 s per trial. Write the one-line reply you would send.

## Part IV

# Pillar II — Differentiable Hardware

# Chapter 10

## Continuous hardware, discrete silicon

The structural theorem of this chapter is corpus-safe (C19) — not a finished lecture. A plain-English restatement lives in `directives/theorem-1-plain.md`; a Swift-free check is `python3 Scripts/tensorlut_math_ref.py`, and the smallest worked example of the LUT view is `python3 Scripts/toy_cipher_demo.py` (INTRO.md). The separable melt-freeze-snap certificate is corpus-safe (C44). Shatter / hold grades and campaign empirics remain seminar-depth and are *not* the theorem [9].

**This section grows when.** A completeness theorem for *multi-LUT topological* melt: that melt recovers cryptographic structure combinatoric sieves miss, for a stated class of netlists beyond a fully observed 1-LUT. Theorem 1'' (C44) is the separable interpolant, not that class. Stream-cipher melts remain on the trajectory.

### 10.1 The problem boolean search cannot see

A combinatoric sieve on a cipher netlist proposes discrete genotypes: INIT bits, plugboard pairs, rotor orders. On a short ciphertext the search space is both enormous and full of linguistic hallucinations (the P1030680 campaign is the case study, not the claim). Boolean search *rejects*. It does not *propose* a nearby reciprocal structure that is almost a stecker.

Pillar II melts the hardware [3, 9].

### 10.2 Setup

Let  $L$  be the number of LUT6 cells and  $w \in [0, 1]^{64L}$  the concatenated INIT genome. Let  $y(w)$  be the *multilinear extension* of those INITs on  $[0, 1]$  (exact on  $\{0, 1\}$  inputs). Let  $t$  be a target bit vector of matching width. Soft crypto fitness and discreteness penalty:

$$F_{\text{crypto}}(w) = -\|y(w) - t\|_2^2, \quad (10.1)$$

$$\pi(w) = \sum_i w_i(1 - w_i), \quad (10.2)$$

$$F(w) = F_{\text{crypto}}(w) - \lambda \pi(w), \quad \lambda \geq 0. \quad (10.3)$$

*Remark 10.1* (Implementation schedule). HELUT's GA uses a squeeze  $\lambda = \lambda_{\max}(p')^2$ . Theorem 1 needs only  $\lambda \geq 0$ .

Emitter:  $E(w)_i = \mathbf{1}[w_i \geq \frac{1}{2}]$ . Freeze mask  $M \subseteq \{1, \dots, 64L\}$  removes frozen coordinates from  $\pi$ . Stecker genotypes are *partial involutions* (disjoint pairs) on  $\{0, \dots, 25\}$ .

Every Yosys \$lut of width 1...6 pads to a 64-wide Float32 INIT. The forward pass is branchless and GPU-shaped. It is not torus FHE.

## 10.3 Theorem 1 (structural)

**Reproduced (C19).** TensorLUT continuous→discrete Theorem 1: six machine-checked lemmas (`TensorLUTFormal.certificate()`). Not a cryptanalytic break. Statement: `directives/tensorlut-theorem.md`. Reproduce: `swift test -c release -filter testTensorLUTFormalCertificate`.

**Theorem 10.1** (TensorLUT continuous→discrete; corpus Theorem 1). *Assume the hypotheses on the certificate: the LUT is multilinear; the GA mutates only unfrozen  $w_i \in [0, 1]$ ; the involution sandwich freezes core INITs. Then:*

1. **Discreteness.**  $\pi(w) \geq 0$ , with equality iff  $w \in \{0, 1\}^{64L}$ .
2. **Crypto MSE.**  $F_{\text{crypto}}(w) \leq 0$ , with equality iff  $y(w) = t$ .
3. **Combined objective.**  $F(w) \leq F_{\text{crypto}}(w)$ ; if  $\pi(w) > 0$ , increasing  $\lambda$  strictly decreases  $F$ .
4. **Emitter.**  $E$  is idempotent on  $\{0, 1\}^{64L}$  and agrees with threshold  $\frac{1}{2}$ .
5. **Involution sandwich.** Overlapping pairs are rejected; applying a valid pair-set twice is the identity on the alphabet.
6. **Freeze.** Coordinates in  $M$  do not contribute to  $\pi$ .

**Proof (machine-checked).** Each clause is a `TensorLUTFormal.check*` in `TensorLUTFormal.swift`, aggregated by `TensorLUTFormal.certificate()`.

**Reproduced (C25).** Theorem 1 corollary: emitter–discrete agreement ( $\pi = 0 \Rightarrow E(w)$  recovers INIT bits) and involution under freeze (`mutatedPreserving` retains frozen pairs). `TensorLUTFormal.corollaryCertificate()`. Reproduce: `swift test -c release -filter testTensorLUTFormalCorollaryCertificate`.

**Theorem 10.2** (TensorLUT emitter / freeze corollary; corpus Theorem 1 corollary). *Under Theorem 10.1 hypotheses plus emit-via-E and mutatedPreserving: if  $\pi(w) = 0$  then  $E(w)$  recovers the binary INIT; every freeze-preserving genotype remains a partial involution containing the frozen pairs.*

Still not melt completeness for arbitrary netlists. The named lemmas of Theorem 10.1 are `discretenessPenalty`, `cryptoFitnessMSE`, `combinedObjective`, `emitterThreshold`, `involutionSandwich`, `freezeMask`. Source path: `Sources/HELUTCore/TensorLUTFormal.swift`.

*Honesty note 10.1.* Theorem 1 does *not* prove recovery of arbitrary keys; a U-534 / P1030680 plaintext; or that melt is complete for all netlists. Shatter / hold grades remain empirical (C8, C9, H6).

**Reproduced (C44).** Theorem 1'' melt-freeze-snap on a coordinate-separable Boolean interpolant: unique maximizer  $w = t$ , snap basin  $|w_i - t_i| < 1/2$ , wrong freeze blocks  $F = 0$ . `TensorLUTFormal.meltFreezeSnapCertificate()`. Reproduce: `swift test -c release -filter testTensorLUTMeltFreezeSnapCertificate`. Not multi-LUT topological melt.

**Reproduced (C48).** Two-LUT cascade  $y = (a \wedge b) \oplus c$ : melt from 0.5, snap, emit two LUT6 Verilog cells. Snapped INIT matches all 8 Boolean corners (a dual interpolant such as NAND+XNOR is allowed). `testTwoLUTCascadeMeltFreezeSnapEmit`. Not Yosys re-synth.

**Theorem 10.3** (TensorLUT melt-freeze-snap, separable interpolant; corpus Theorem 1''). *Under a fully observed 1-LUT (each used INIT address an independent Boolean target  $t$ ),  $F(w) = -\|w - t\|_2^2 - \lambda\pi(w)$  has unique maximizer  $w = t$ ; the emitter recovers  $t$  on the open cube  $|w_i - t_i| < 1/2$ ; a freeze away from  $t$  makes  $F = 0$  unreachable.*

## 10.4 Five-cell test for Theorem 1

| Cell        | Artifact                                                                                  |
|-------------|-------------------------------------------------------------------------------------------|
| Proof       | Theorem 10.1 + <code>TensorLUTFormalCertificate</code>                                    |
| Table       | Lemmas vs holds on the certificate; campaign grades in <code>BREAK_P1030680.md</code> §21 |
| Metric      | $F_{\text{crypto}} = 0$ on unmutated baseline; $\pi = 0$ on binary INIT                   |
| Examples    | M4 baseline emit; freeze-core involution; blind 3-pair PASS                               |
| Application | Stecker search that cannot propose a non-reciprocal map                                   |

**Exercise 10.1** (Read a certificate). Run the reproduce filter (Appendix D) or, on the theory track, read `TensorLUTFormal.certificate()` and list the six holds bits. For each lemma, write one sentence a journalist may print and one they may not.

## 10.5 Shatter versus hold (empirical)

A full cold-start on a large sequential cipher discovers continuous shortcuts that *shatter* when  $\lambda$  rises:  $F_{\text{crypto}}$  looks perfect in float and dies on the Boolean cube. Targeted melting freezes known-good LUT blocks (`freezeMask`) and evolves only the unknown cone—clause 6 of Theorem 10.1 is why frozen coordinates do not pay  $\pi$ . Early resignation reboots doomed lineages rather than squeezing a reciprocal shortcut to the end.

**Definition 10.1** (Grade). Under a published  $\lambda$  schedule, a lineage **holds** if the emitted binary netlist preserves  $F_{\text{crypto}} = 0$  (or a stated bar). It **shatters** if discreteness kills the continuous solution. Both grades are science. Shatter is not a failed demo; it is evidence about shortcuts. It is *not* a clause of Theorem 10.1.

**Reproduced (C8, C9).** TensorLUT M4 baseline  $F_{\text{crypto}} = 0$  plus Verilog emit; Stecker involution sandwich with blind 3-pair PASS. Method claims, not a U-534 / P1030680 plaintext.

## 10.6 Involution sandwich

Stecker genotypes are partial involutions (disjoint pairs) by construction (Theorem 10.1, clause 5). The GA cannot propose a non-reciprocal map. The 16 cone LUTs around an identity plugboard cannot invent letter swaps by INIT melt; the sandwich injects  $S(\text{CT})$  into a frozen core and scores soft PT against bits of  $S(P)$ . Reciprocity is structural.

*Honesty note* 10.2. Short cribs leave unused pairs unconstrained and can admit false  $F = 0$  steckers. Grade active-map agreement on  $\text{CT} \cup \text{PT}$ , use parsimony, grow the pair budget on plateau, and allow soft freeze + thaw. TensorLUT grades are not a campaign decrypt (**N**, **H6**).

## 10.7 Adversarial synthesis as a compiler loop

Forward: silicon  $\rightarrow$  uniform float tensors. Reverse: cooled tensors  $\rightarrow$  INIT hex  $\rightarrow$  Verilog LUT instantiations. The loop is a generative compiler in *this* laboratory: cooled tensors emit LUT6 Verilog, not a bitstream onto a physical FPGA. Theorem 10.1 is the invariant the loop is forced to obey. It is not a guarantee that the loop finds a key.

**Exercise 10.2.** Why must  $\pi(w) = \sum w_i(1 - w_i)$  vanish on binary INITs, and why is a fitness that ignores  $\pi$  allowed to “solve” a cipher that no FPGA can implement? Give a two-LUT numerical cartoon. Which clause of Theorem 10.1 did you just unpack?

**Exercise 10.3** (Seminar). Theorem 10.1 is structural, not complete. Write the statement of a completeness theorem you wish were true (melt recovers  $X$  on netlist class  $Y$  under schedule  $Z$ ), and the counterexample shape you would try first. Do not upgrade shatter/hold grades into that theorem.

**Non-claim.** Pillar II does not break P1030680, does not replace Welchman search, and does not make TensorLUT a PBS. Parallel research; campaign fitness stays cleartext. **C19** is a theorem about the objective and the involution contract, not about plaintext.

## Part V

# Pillar III — Adversarial Polymorphic Ciphers

# Chapter 11

## Ciphers that grow under fire

**This section grows when.** A polymorphic Red/Blue standard beyond the bounded E256 fixture-v4 receipt (roadmap: Schneier pillar), and IND-CPA / Red-pressure security results that go past finite structural checks. This chapter is a scaffold and collecting surface for existing receipts—not lecture-ready material and not a finished industry standard.

### 11.1 Evidence boundary for a design laboratory

Historical Enigma traits motivate the E256 design laboratory, but the checked claims here do not prove those traits universally fatal or prove their removal for every generated machine. This collecting surface records three existing rows: bounded fixture-v4 implementation parity (C10), a bounded structural certificate / protocol (C24), and a historical encrypted teaching cone (C39).

The same-state reciprocal shape is a design target. The evidence below is finite: only checks explicitly called exhaustive are exhaustive, and key/state coverage remains bounded.

**Open E256-H/X0 hardware and policy measurements.** The E256-061/062/063 hardware and X0 checks remain OPEN\_PROGRESS and have no C/H/N row. Their complete dependency-ordered reproduction commands, receipt digests, and non-claims are in REPRODUCE.md. The three latest gates can be re-checked directly with:

- `make e256-hw-h2-check;`
- `make e256-hw-h4-sequence-check;`
- `make e256-x0-h4-evolution-check.`

The H4 census validly found that 737,280 of 1,474,560 preregistered mixed three-round windows fail the exact all-to-all byte-dependency criterion. Its first frozen-order witness uses list indices [1, 2, 3]—tuples (0, 1, 3, 4), (0, 1, 4, 3), and (0, 1, 4, 7)—where output lane 0 lacks input lanes [6, 11, 24, 29]. The sequence-specific H3 collapse passed 442,368 exact comparisons; integrated RTL/model and per-input cycle evidence passed 55,296 comparisons each; and all 19 controls were detected.

X0 then checked the exact criterion  $S(y) + S(z) = \mathbb{Z}_8$  for all 147,456 ordered catalog pairs using independent literal and sumset implementations. Exactly 73,728 directed transitions are safe and every node has in/out degree 192. The full graph has two 192-node strongly connected components; the 16-support quotient has two 8-node components and no Hamiltonian cycle. The frozen fallback certificate alternates catalog indices 0 and 3 for twelve selectors and verifies all 11 adjacent transitions. All 23 X0 controls were detected. This proves only the represented contiguous three-round structural dependency windows; the two components prevent one safe walk from covering all 16 supports.

The generic Yosys/ABC sequence-minus-repeated observations were:

- FF storage:  $-407$  LUT6,  $+132$  state bits,  $-275$  primitive cells, and  $+1$  combinational level;
- ring storage:  $-57$  LUT6,  $+132$  state bits,  $+75$  primitive cells, and  $+1$  combinational level.

There was no threshold or winner. Negative LUT deltas from separate generic mappings are not physical selector savings. The H4 counterexample parks unrestricted arithmetic switching; X0 supplies a bounded safe-transition grammar and path, not a production evolution mechanism. Neither result inherits the fixed-tuple four-round/25-active-S-box statement. These are generic mapped-RTL measurements, bounded falsification, and host-side structural evidence only—not a theorem or course-ready laboratory result. They select no architecture, storage organization, round count, schedule, XOF, H4 policy, suite, protocol, production RTL, cryptanalytic result, or security value.

## 11.2 Fixture-v4 evidence capsule (scaffold)

- The fixed live profile is schema-4 E256-v2/gen0, fixture-v4.
- Its conjugated-XOR center is  $A_i^{-1}(A_i(x) \oplus k_i)$ .
- The host derives and transports (payload, centerMask, absoluteByteCounter); RTL validates absoluteByteCounter and does not implement HMAC.
- The formal C24 test is **1/1 PASS**; the full E256 C10 suite is **49/49 PASS**. The receipt is logs/e256-v2-gen0-fixture-v4-validation.json.

**Reproduced (C10).** Enigma256 fixture-v4 bounded implementation / KAT-parity, reciprocity, and conjugated-XOR evidence for the fixed schema-4 E256-v2/gen0 native profile. Full E256 suite: **49/49 PASS**. Receipt: logs/e256-v2-gen0-fixture-v4-validation.json.

**Reproduced (C24).** Enigma256 SoftBus Theorem 2 is a bounded fixture-v4 structural certificate / protocol: deterministic four-state scramble-bijection sweep; exhaustive byte reciprocity for one frozen fixture state; deterministic 128-byte stream round-trip; fixture plugboard plus selected XOR-center involution/fixed-point checks; and exact schema-4 E256-v2/gen0 native-profile integrity. Formal test: **1/1 PASS**. Statement: directives/enigma256-theorem.md.

**Reproduced (C39).** One frozen historical one-byte / one-round E256 teaching cone with algebraic sboxes, legacy UKW, offsets 0, and identity plug under GGSW Metal SING at  $N=1024$ . It predates the live fixture-v4 profile and does not exercise transported mask/counter tuples, the conjugated-XOR center, live BRAM day-key tables, NLFF stepping, or the full core.

## 11.3 Theorem 2 (bounded structural certificate / protocol)

**Theorem 11.1** (E256 fixture-v4 bounded structural certificate / protocol; corpus Theorem 2). *For the fixed schema-4 E256-v2/gen0 native-profile fixture-v4, the C24 bounded structural certificate / protocol checks exactly: (i) a deterministic four-state scramble-bijection sweep; (ii) exhaustive*

byte reciprocity for one frozen fixture state; (iii) a deterministic 128-byte stream round-trip; (iv) the fixture plugboard plus selected XOR-center involution/fixed-point checks, where the center is  $A_i^{-1}(A_i(x) \oplus k_i)$ ; and (v) exact schema-4 E256-v2/gen0 native-profile integrity. The host derives and transports (*payload*, *centerMask*, *absoluteByteCounter*). RTL validates *absoluteByteCounter* and does not implement HMAC.

Machine-checked result: `Enigma256Formal.certificate()` / `testEnigma256FormalCertificate`, **1/1 PASS**; full E256 suite, **49/49 PASS**. Receipt: `logs/e256-v2-gen0-fixture-v4-validation.json`. Reproduce through `REPRODUCE.md`.

*Honesty note* 11.1. Theorem 2 records a *bounded structural certificate / protocol*, not a universal proof. Finite checks are exhaustive only where stated, and state/key coverage is bounded. It is not IND-CPA, an HMAC proof, external cryptanalysis, human acceptance, or a finished polymorphic-cipher standard.

## 11.4 Red-pressure questions kept separate

TensorLUT, known-plaintext, and `ent` experiments remain internal Red questions on this collecting surface. C24 does not show that those experiments cannot force a generation roll, and the fixture-v4 receipt is not external cryptanalysis or human acceptance. No coupled-NLFF hardening result is part of the bounded certificate.

## 11.5 Draft lab prompts (scaffold; not lecture-ready)

**Exercise 11.1** (Blue studio prompt). Propose one additional E256 invariant as an open design question (for example, an NLFF period lower bound or BRAM burst-width condition). Specify a test that could refuse a generation, but do not promote the proposal to a claim without a checked receipt.

**Exercise 11.2** (Historical-leak checklist prompt). Compare historical Enigma traits with the corresponding E256 design choices. Keep every unsupported security consequence labeled as a question, not a claim.

## 11.6 Relation to Pillars I and II

This collecting surface links the bounded C24 protocol to its fixture receipt (Theorem 11.1). **C39** is a historical frozen one-round legacy-UKW scramble under GGSW Metal SING. It predates live fixture-v4 and does not exercise transported *centerMask*/*absoluteByteCounter*, the conjugated-XOR center, live BRAM, NLFF, or a full core. Calling that lab “production encrypted E256” is a **N**.

**Non-claim.** E256 is not a NIST candidate in this scaffold, not a replacement for TLS, and not an implication that polymorphic rotors beat lattices. The fixture-v4 receipt is bounded implementation / protocol evidence, not IND-CPA, an HMAC proof, external cryptanalysis, or human acceptance.

# Part VI

## Laboratory, Applications, and the Frontier

# Chapter 12

## Application gallery

Applications exist to keep pillars honest. A compiler with no netlist is a speech; a melt with no cipher is a float toy; a polymorphic cipher with no Red is a blog post.

Canonical nine-slot card: `directives/application-gallery.md`.

### 12.1 Pillar I (encrypted)

1. **Encrypted full\_adder SING.** Multi-LUT correctness at production-shaped  $N$  (**C6**, **C20**; **C21** is  $e=0$  BK). Covering noisy adder is **C52**.
2. **Encrypted tree / regex SING.** Compiler is not adder-specialized (**C6** demo  $N$ ; regex covering **C57** at production  $N$ ). Tree Metal covering remains.
3. **Hardness + noisy-BK certificates.** Estimator fill-in and covering-gadget residuals (**C5**, **C22**, **C23**). Covering Track A at  $N=1024$ ,  $\sigma=128$ ,  $k=7$  is **C52–C54**; cheaper covering-b2 is **C57**. Product-shaped `cryptoPublicMS` inject remains a graded failure (**C26**, **C56**); native  $k=1$  remains **C37/C55** (**H4** remainder).
4. **Encrypted sequential.** Counter (**C38**), toy NOP/ADD (**C40**), PicoRV32 `addi+sw+lw` at demo  $N=8$  (**C50**); Metal NOP-fetch (**C51**).
5. **4-bit CSA vs ripple LUT cut (C42).** Architecture, not TensorLUT melt.

### 12.2 Pillar I shape laboratory (oracle — not FHE)

1. **Decision tree.** A 4-bit non-linear boundary, 7 LUTs. Exact threshold logic on a batch of records. Teaches: non-linear Boolean  $\neq$  neural net, and  $B$  is a capacity knob.
2. **Regex / DFA search.** 3-character matcher, 23 LUTs. Large  $B$  turns unified memory into the story.
3. **PicoRV32.**  $\sim 4.8k$  LUTs,  $\sim 1.5k$  DFFs. Scripted `reseth` boot. Oracle path: **C1**. Encrypted 1-tick at demo  $N=8$ : **C50** `lw` sees 1; Metal NOP-fetch **C51**.

## 12.3 Pillar II

1. M4 TensorLUT baseline emit (C8).
2. Stecker involution sandwich / formal (C9, C19, C25, C44, C48).
3. Shatter vs hold under  $\lambda$  — seminar empirics; not a decrypt (H6).

## 12.4 Pillar III

1. SoftBus reciprocity / bijection (C10, C24).
2. Encrypted 1-round frozen scramble (C39) — algebraic slice under torus FHE, not live BRAM.
3. Red battery (TensorLUT / KPA / ent) — empirical grades in logs/enigma256-\*.
4. Fail-closed NLFF harden (C24 clause 5).

## 12.5 Campaign as a case study (not a decrypt)

Welchman-path break of known P1030684 (C2),  $\leq 10$ -plug SAT kill chain (C3), Potsdam/Plaice  $\neq$  P1030680 (C11), windowed discriminator versus whole-message turnover (C12): these belong in a cryptanalysis seminar attached to Pillar II methods.

**Non-claim.** This book does not claim a decrypt of P1030680. Campaign fitness is cleartext Metal batch, not HELUT encrypted tick rate.

**Exercise 12.1.** Pick one application tier and write a five-cell card for it. If you cannot name two runnable instances, it is not ready for the gallery lecture.

**This section grows when.** An encrypted tree/regex Metal SING table at production  $N$ ; PicoRV covering at production  $N$ ; any new application that earns a **C** row. Site gallery figures already exist under `site/public/gallery/`.

# Chapter 13

## Laboratories

Systems-track labs assume macOS 14+, Apple Silicon, Swift 6.3, and a release `helut` at repo root: `swift build -c release -product helut`. Theory-track substitutes are stated per lab.

### 13.1 Lab 0 — Claim literacy (everyone)

Read `directives/claim-sheet.md` at the printed epoch. Classify five public sentences (instructor-provided) as **C**, **H**, **N**, or frontier. No machine required.

### 13.2 Lab 1 — Negacyclic oracle

Implement  $N = 8$  schoolbook mul in  $\mathbb{Z}/2^{32}\mathbb{Z}[X]/(X^N + 1)$  and check a dense Toeplitz expansion against it on torus corners. Theory track:  $N = 4$  by hand (Chapter 6 exercises).

### 13.3 Lab 2 — Netlist clock

Compile the checked-in `counter` or `full_adder` Yosys JSON from `Generated/` with `-compile-only`. If you re-synthesize an authored source from `Hardware/`, write the disposable JSON below `build/hardware/`; do not overwrite a root compatibility file. Tick with `-bench`. Identify  $B$ ,  $N$ , LUT count, DFF count. Explain in writing why  $N$  is not the circuit size.

### 13.4 Lab 3 — Boolean oracle versus FHE path

```
.build/release/helut --bench Generated/Netlists/Examples/netlist.json --degree 64 \  
  --lut-backend pbs --encoding phase --bench-equiv  
.build/release/helut --bench Generated/Netlists/Examples/netlist.json --degree 8 \  
  --bench-encrypted --cpu-only --sing --vectors 8  
.build/release/helut --bench Generated/Netlists/Examples/counter_netlist.json --degree 8  
  \  
  --bench-encrypted --cpu-only --sing --vectors 8  
.build/release/helut --bench Generated/Netlists/Examples/toy_isa_netlist.json --degree 8  
  \  
  --bench-encrypted --cpu-only --sing --vectors 32  
.build/release/helut --bench Generated/Netlists/PicoRV32/picorv32_netlist.json --degree 8  
  \  
  
```

```
--bench-encrypted --cpu-only --sing --ticks 48 --reset-hold 3 \  
--encrypted-mem prog --paths 'blind-rotate public-ms boolean'
```

Deliverable: a table with columns {backend,  $N$ , SING, certificates printed, may we say “encrypted” on a slide?}.

## 13.5 Lab 4 — Production-shaped SING

**Reproduced (C6).** full\_adder encrypted SING through  $N = 1024$  (the result that closed **H2**).

Follow Appendix D. Record ms/row at  $N = 8$  and  $N = 1024$ . Write the **H1** asterisk next to hardness. Write the **H4** asterisk next to depth: covering Track A Grade B is **C52–C54**;  $e=0$  BK is still the default noiseless SING; native  $k=1$  / cryptoPublicMS / PicoRV covering remain.

Theory track: parse a published SING log and reconstruct the table without running Metal.

## 13.6 Lab 4b — Measured noisy BK (C22)

```
.build/release/helut --measure-bk-noise --degree 8 \  
--trials 8 --bk-noise 64
```

Deliverable:  $B_{\text{bk}}$ ,  $\hat{\sigma}$ ,  $\delta/2$ , and one sentence on why  $\ell = 1$  booleanPublicMS is omitted. Theory track: read logs/helut-noisy-bk-measure.log. Optional: parse a **C52** covering log at  $N=1024$  and name the three **H4** remainders (**C37/C55**, **C26/C56**, PicoRV covering).

## 13.7 Lab 5 — Metal compiler post-mortem

Compare fused-megagraph DNF, **C16–C21** using Chapter 8’s table. Optional systems: make test-metal-p1 and one -bench-encrypted-micro -degree 64 run. Deliverable: the **H3** paragraph in your own words (scheduler vs NTT).

## 13.8 Lab 6 — TensorLUT Theorem 1 (C19)

```
swift test -c release --filter testTensorLUTFormalCertificate
```

Deliverable: name the six lemmas, the certificate hypotheses, and one sentence that Theorem 10.1 does *not* prove. Optional: testTensorLUTMeltFreezeSnapCertificate (**C44**) — what class of INIT genomes the unique-maximizer lemma covers. Theory track, no Swift and no Mac: python3 Scripts/toy\_cipher\_demo.py for the smallest worked example, then python3 Scripts/tensorlut\_math\_ref.py; read directives/theorem-1-plain.md and directives/tensorlut-theorem. reconstruct the five-cell card. Both scripts are executable checks, not machine-checked proofs.

## 13.9 Lab 7 — Five-cell audit (everyone)

The signature assignment (Chapter 2). One abstract sentence from paper/helut.tex; five cells; a pass/fail on whether the sentence is lecture-safe at this epoch.

## 13.10 Lab 8 — Frontier proposal (everyone)

Chapter 15: pick one avenue. Propose the first experiment that would mint a **C** row. You may not claim the avenue works.

## 13.11 Lab 9 — Run it five times

Every measurement in this corpus began as a single observation. This lab is about what that costs, and it needs no Apple hardware for the first half.

1. Pick any timing in `directives/claim-sheet.md`. Re-run it five times with `Scripts/bench_repeat.py` and report min, median, max and relative spread. Was the archived figure a median, or a best case?
2. Run the same command once on a cold cache and once warm. On the machine these claims were measured, one row moved by  $2.4\times$  between the two. Which of the two numbers belongs in a paper?
3. Now take a row whose claim is a *failure*. Run it five times. Report whether the failure is deterministic. A negative that reproduces three times in five is a different scientific object from one that reproduces five times in five — name the difference in one sentence.
4. Read `logs/leeloo/prefix-vs-postfix.md`. Two rows changed stability after a one-line source fix. Explain why running each row once could never have found that, no matter how carefully the code was reviewed.

Deliverable: a table of {row, archived figure, median of 5, spread, verdict} and one paragraph on which archived numbers you would now decline to quote.

*Honesty note* 13.1. The harness refuses to start if another benchmark is already running, and that refusal is load-bearing rather than fussy: one benchmark on a sixteen-core machine is about 26% per core, well under any sane load threshold, yet it contends for the GPU and memory bandwidth and will skew every number in a concurrent sweep. A build is worse — it replaces the binary underneath a running sweep.

**This section grows when.** Negative-stability results exist for the **C60/C61** rows, so students can compare a deterministic negative against an intermittent one on real data.

## 13.12 Optional project slots

TensorLUT involution sandwich replay (**C9** logs), E256 invariant studio, or a new Yosys netlist (student RTL  $\rightarrow$  SING). Instructor approval required; campaign decrypt projects are out of scope.

**This section grows when.** Autograded SING harnesses for course VMs; a Dockerized theory track; non-Apple ring-oracle labs.

## Chapter 14

# Open problems (trajectory)

This chapter is the discovery path after disclosure [8]. None of these items are lecture facts until they have receipts.

## 14.1 Near term

| Track                             | Status at epoch 2026-09-23 / C69                                                                                                                                            | Next experiment                                                                                                                                                                                                                                                                                               |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>H2</b> full_adder $N \geq 256$ | <b>Closed</b> ( $Z_{2N}$ pack / rotationPower)                                                                                                                              | Keep as a worked bug in Chapter 6.                                                                                                                                                                                                                                                                            |
| <b>H1</b> Sage lattice-estimator  | <b>C23</b> filled; production $ \Delta  = 4.5$ ; core-SVP vs Cost rop divergences                                                                                           | Optional retune / quote estimator-only on $\Delta > 16$ rows.                                                                                                                                                                                                                                                 |
| <b>H3</b> Metal BR at large $N$   | <b>C20</b> boolean 10.6 s; <b>C21</b> crypto $\ell = 2$ 11.38 s                                                                                                             | NTT inside crypto $\ell = 2$ at $N=1024$ (incomplete public-MS gadget).                                                                                                                                                                                                                                       |
| <b>H4</b> Noisy BK                | <b>C52–C54</b> covering-b1; <b>C57</b> covering-b2 $\varepsilon + \text{regex SING @ } N=1024$ ; covering KS reaches $n=512$ ( <b>C69</b> , the old $n=512$ FAIL withdrawn) | Native $k=1$ ( <b>C37/C55</b> ); cryptoPublicMS ( <b>C26/C56</b> ); PicoRV covering ( <b>C60</b> Q FAIL).                                                                                                                                                                                                     |
| Encrypted sequential              | <b>C53/C54</b> covering-b1 counter + toy ISA @ $N=1024$ $\sigma=128$ $k=7$ ; <b>C51</b> PicoRV Metal still demo $N=8$                                                       | Metal PicoRV covering @ $N=1024$ (LUT-tax).                                                                                                                                                                                                                                                                   |
| Campaign catalog                  | Middle ring $\neq A$ tested under VIII-fast (24/24 dead; 294 WO open); catalog parked @417                                                                                  | Resume -bombe-from 418.                                                                                                                                                                                                                                                                                       |
| Garble / quarantine               | Soft-band grades                                                                                                                                                            | Sister-message lessons; not a decrypt claim.                                                                                                                                                                                                                                                                  |
| Audience R1–R6                    | Shipped, not <b>C</b> rows: Linux checks, weak-vs-affine toy cipher pair, Theorem 1 in English, the $q$ -split, four-page note                                              | Open: Lean/Coq version of the six clauses; toy cipher through Yosys; $q = 2$ FHE.                                                                                                                                                                                                                             |
| Encrypted determinism             | Fixed 2026-08-15: input encryption walked a Dictionary while drawing from the shared RNG, so ciphertexts varied per process                                                 | End-to-end determinism assertion in CI (byte-identical ciphertexts across processes at a fixed seed), not just the in-process guard (§3.7).                                                                                                                                                                   |
| $\varepsilon$ sample efficiency   | <b>C22</b> resolved at $N=128$ ( $-18.0$ , 95% bound $-16.1$ , $n=512$ ); $N=1024$ costs 155 s/trial so $n \approx 8450$ is $\sim 15$ days                                  | Sample all $N$ accumulator coefficients instead of the one extracted residual ( $\sim 1000 \times$ more samples per bootstrap). Until then <b>C35</b> 's bar is supportable only at inject $B \leq 4$ (§9.6).                                                                                                 |
| Vertex maximizers                 | Literature-resolved, not a <b>C</b> row                                                                                                                                     | Raising $\lambda$ is the classical concave exact penalty for 0–1 programming (Raghavachari 1969; Giannessi–Niccolucci 1976). Sufficient: $\lambda \geq \frac{1}{2} \sup \lambda_{\max}(\nabla^2 f)$ , measured $2 + \sqrt{3}$ on two LUTs. Open: scaling with depth / fan-out. note/lut-relaxation.tex §4–§5. |

## 14.2 Mid-term pillars

1. **Pillar I science:** shallower nets, NTT/persist graphs, estimator-backed params (this book's Parts III).
2. **Pillar II science:** Theorem 1 landed (**C19**); corollary emitter / freeze (**C25**); melt-freeze-snap on a separable interpolant (**C44**); 2-LUT cascade emit (**C48**); remaining: multi-LUT topological

melt, stream-cipher melts.

3. **Pillar III science:** SoftBus Theorem 2 landed (**C24**, Chapter 11); remaining: polymorphic Red/Blue standard beyond E256 SoftBus.
4. **FHE gate / ZK depth:** TensorLUT aimed at multiplicative depth (queued; not a part of this edition).
5. **Side-channel (parked):** bgpucap-style power on live Metal graphs only with controlled fixtures — Chapter 15, avenue 5.

**Exercise 14.1** (Pick a next experiment). Choose one open hedge. Write a one-page lab proposal: command line, pass bar, which box in this book would change, and which **N** you refuse to imply if the experiment fails.

**This section grows when.** Each closed row in this table deletes a hedge box elsewhere and adds a reproduced box. That is the only way this chapter shrinks.

# Chapter 15

## Frontier (not claims)

The text in this chapter is adapted from the project’s speculative avenues [6]. It is included so the public story stays honest: **reproducible core now; weird frontier labeled as frontier.**

Instructors: last two seminars, or a 10% essay. Never a true/false bank.

### 15.1 Encrypted LLM guardrails

**Frontier — not a claim.** Evolve a tiny neural or logic-gate firewall in the encrypted domain that sits between an untrusted prompt and a local model. The firewall circuit would evaluate in the dark on token tensors and mutate its Boolean topology from adversarial inputs it blocked—without decrypting prompt, weights, or mutation rules in memory.

First graduating experiment (suggested): a 4-LUT encrypted classifier SING on synthetic token bits, with a certificate, *before* any talk of genetic mutation or Ollama.

### 15.2 Self-modifying encrypted binaries

**Frontier — not a claim.** A PicoRV32 that already ticks on an **MPSGraph** under oracle encodings (**C1**) is not a dark VM. Encrypted PicoRV is demo  $N=8$  at 4.0 bits (**C45–C51**), not production  $N$ . The avenue is a binary that rewrites its own instructions while remaining FHE-encrypted, so the host sees tensor multiplies and cannot inspect the guest.

First graduating experiment: encrypted instruction-fetch SING on a toy ISA smaller than PicoRV32, with a written ethics note. This is a privacy architecture problem, not a malware homework.

### 15.3 Honey-token ledgers

**Frontier — not a claim.** Batched encrypted pattern search composed with a zero-knowledge ledger, so a sync evaluates trapdoor circuits without the storage node learning which file or which trigger fired.

First graduating experiment: regex netlist SING at modest  $B$  with an explicit non-claim that the node learns nothing—then measure what the *host process* still sees (lengths, timings). Honesty before mesh networks.

## 15.4 Genetic hardware synthesis

**Frontier — not a claim.** `fhe-evolve` mutates truth tables and connectivity; Yosys recompiles; MPSGraph re-lowers; the running system processes a stream through hardware it just invented.

First graduating experiment: one offline generation of a 2-LUT netlist that SING-matches a fixed spec, wall-clock of synth+lower, no on-the-fly claims.

## 15.5 Metal-tick power analysis (parked)

**Frontier — not a claim.** Treat the live Metal FHE pipeline as the device under test. Capture GPU-side power only while a tick-synchronized graph runs *controlled fixtures*. Classical DPA/CPA against aligned traces. A distinguish or a clean negative are both science. Opportunistic capture of arbitrary workloads is out of scope.

First graduating experiment: fixture harness that emits tick markers and a published “no analysis yet” log. Do not assign attacks against third-party graphs.

## 15.6 Rules for frontier essays

1. Label every paragraph frontier or first-experiment.
2. Name the **C** row it would become, and the **N** it must not imply on the way.
3. Cite Chapter 3. If the five cells cannot even be *imagined*, the essay is not ready.

# Part VII

## Appendices

# Appendix A

## Notation

| Symbol            | Meaning                                                                                                                                                                   |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $q$               | Torus modulus; HELUT uses $q = 2^{32}$ .                                                                                                                                  |
| $N$               | Polynomial degree / ciphertext shape. Not LUT count.                                                                                                                      |
| $n$               | LWE dimension (often $n = N$ in HELUT boolean production).                                                                                                                |
| $B$               | Batch axis: independent copies of one netlist.                                                                                                                            |
| $\delta$          | $q/(2N)$ , rotation / message spacing.                                                                                                                                    |
| $\sigma$          | LWE Gaussian width.                                                                                                                                                       |
| $\varepsilon$     | Ingest failure probability (Gaussian certificate).                                                                                                                        |
| $B_{\text{bk}}$   | BK noise bound after BR; noiseless (0) on default Track A throughput SING; covering Track A Grade B is <b>C52–C54 (H4)</b> , also <b>C22/C36</b> .                        |
| $\ell$            | GGSW gadget levels.                                                                                                                                                       |
| $W$               | CMUX tile width in the Metal compiler.                                                                                                                                    |
| $R_q$             | $\mathbb{Z}_q[X]/(X^N + 1)$ .                                                                                                                                             |
| $M_A$             | Negacyclic Toeplitz matrix of polynomial $A$ .                                                                                                                            |
| $w$               | TensorLUT INIT genome in $[0, 1]^{64L}$ .                                                                                                                                 |
| $L$               | Number of LUT6 cells in a TensorLUT netlist.                                                                                                                              |
| $y(w)$            | Multilinear extension of INITs (exact on $\{0, 1\}$ ).                                                                                                                    |
| $t$               | Target bit vector for $F_{\text{crypto}}$ .                                                                                                                               |
| $\pi(w)$          | Discreteness penalty $\sum_i w_i(1 - w_i)$ .                                                                                                                              |
| $E(w)$            | Emitter $\mathbf{1}[w \geq \frac{1}{2}]$ .                                                                                                                                |
| $M$               | Freeze mask: coordinates excluded from $\pi$ .                                                                                                                            |
| $\lambda$         | Discreteness squeeze on $\pi(w)$ ; Theorem 1 needs $\lambda \geq 0$ .                                                                                                     |
| <b>\$lut</b>      | Yosys look-up cell; one programmable gate.                                                                                                                                |
| SING              | <b>Stimuli In, Golden out</b> : encrypted evaluation $\equiv$ cleartext netlist on the same generic input vectors ( <b>-sing</b> ); certificate lines ride with the tick. |
| <b>Cn, Hn, Nn</b> | Claim, hedge, non-implication IDs.                                                                                                                                        |

## Appendix B

# Claim index (snapshot)

Canonical living inventory: `directives/claim-sheet.md`. If this appendix disagrees with the sheet, the sheet wins. Snapshot epoch: 2026-09-23 / C69.

### B.1 Reproducible results

| ID         | Result                                                                                                                                                                                                                                                                                                                                                                                |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C1</b>  | Yosys \$lut/DFF → Metal/CPU tensor eval (oracle).                                                                                                                                                                                                                                                                                                                                     |
| <b>C2</b>  | Welchman path breaks known P1030684 end-to-end.                                                                                                                                                                                                                                                                                                                                       |
| <b>C3</b>  | ≤ 10-plug / SAT kill chain removes ghosts.                                                                                                                                                                                                                                                                                                                                            |
| <b>C4</b>  | Encrypted LWE/GLWE + GGSW BR per \$lut.                                                                                                                                                                                                                                                                                                                                               |
| <b>C5</b>  | Certificates on encrypted ticks.                                                                                                                                                                                                                                                                                                                                                      |
| <b>C6</b>  | Encrypted ≡ clear multi-netlist SING; adder $N \leq 1024$ .                                                                                                                                                                                                                                                                                                                           |
| <b>C7</b>  | Metal 1-LUT BR microbench (fused ancestor numbers).                                                                                                                                                                                                                                                                                                                                   |
| <b>C8</b>  | TensorLUT M4 baseline $F_{\text{crypto}} = 0$ + Verilog emit. The restored generic per-netlist CLI regenerates the 925-LUT / 49-DFF artifact exactly, including all 59,200 INIT floats. Emitted Verilog re-synthesizes through Yosys and is functionally equivalent on the 4-LUT design over all 16 inputs, with a one-INIT-bit negative control.                                     |
| <b>C9</b>  | Stecker involution sandwich; blind 3-pair PASS.                                                                                                                                                                                                                                                                                                                                       |
| <b>C10</b> | Enigma256 fixture-v4 bounded reciprocity / conjugated-XOR / KAT parity for the schema-4 E256-v2/gen0 native profile; full E256 suite <b>49/49 PASS</b> ; center $A_i^{-1}(A_i(x) \oplus k_i)$ ; receipt <code>logs/e256-v2-gen0-fixture-v4-validation.json</code> . Bounded implementation / parity evidence—not IND-CPA, an HMAC proof, external cryptanalysis, or human acceptance. |
| <b>C11</b> | Potsdam/Plaice keys $\neq$ P1030680.                                                                                                                                                                                                                                                                                                                                                  |
| <b>C12</b> | Windowed discriminator vs whole-message turnover flaw.                                                                                                                                                                                                                                                                                                                                |
| <b>C13</b> | Metal tiled-kernel BR at $N = 1024$ (pre-fusion lineage).                                                                                                                                                                                                                                                                                                                             |
| <b>C14</b> | Metal full_adder SING at $N = 1024$ (pre-NTT lineage).                                                                                                                                                                                                                                                                                                                                |
| <b>C15</b> | Metal netlist-scheduled SING at $N = 1024$ .                                                                                                                                                                                                                                                                                                                                          |
| <b>C16</b> | Metal fused EP kernel; 1.043 s/BR at $N = 1024$ .                                                                                                                                                                                                                                                                                                                                     |
| <b>C17</b> | GPU-resident BR tile; 0.519 s/BR; SING 12.2 s/8.                                                                                                                                                                                                                                                                                                                                      |
| <b>C18</b> | 3-prime NTT persist BR; 0.433 s/BR; SING 15.6 s/8.                                                                                                                                                                                                                                                                                                                                    |
| <b>C19</b> | TensorLUT continuous→discrete Theorem 1 (six machine-checked lemmas).                                                                                                                                                                                                                                                                                                                 |
| <b>C20</b> | Wavefront-parallel independent \$lut BRs; boolean SING 10.6 s/8.                                                                                                                                                                                                                                                                                                                      |
| <b>C21</b> | Metal cryptoPublicMS $\ell = 2$ SING at $N = 1024$ ; 11.38 s/8.                                                                                                                                                                                                                                                                                                                       |

*Continued on next page*

| ID         | Result ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C22</b> | Measured noisy-BK residual $\rightarrow B_{\text{bk}}/\hat{\sigma}$ (covering gadget).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>C23</b> | Native Sage lattice-estimator fill-in; production $ \Delta  = 4.5$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>C24</b> | Enigma256 SoftBus Theorem 2, bounded fixture-v4 structural certificate / protocol: formal <b>1/1 PASS</b> ; deterministic four-state scramble-bijection sweep; exhaustive byte reciprocity for one frozen fixture state; deterministic 128-byte stream round-trip; fixture plugboard plus selected XOR-center involution/fixed-point checks; exact schema-4 E256-v2/gen0 native-profile integrity. Host derives/transport (payload, centerMask, absoluteByteCounter); RTL validates the counter and does not implement HMAC. Finite checks are exhaustive only where stated; state/key coverage is bounded. Full E256 suite <b>49/49 PASS</b> ; receipt logs/e256-v2-gen0-fixture-v4-validation.json. |
| <b>C25</b> | TensorLUT Theorem 1 corollary (emitter-discrete + involution under freeze).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>C26</b> | Noisy-BK identity residual at $N=1024$ : inject $B=64$ undecodable; $B=4$ $\varepsilon \log_2 \approx -1$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>C27</b> | Exact public-MS covering under $q=2^{32}$ only at $N \in \{8, 128\}$ (Theorem 3).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>C28</b> | Track B Metal cryptoPublicMS SING at $N=128$ with noisy BK ( $B=64$ ) PASS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>C29</b> | Exact public-MS covering stays $\{8, 128\}$ for any power-of-two $q=2^w$ (Theorem 3').                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>C30</b> | Track B $\varepsilon$ vs inject $B$ at $N=128$ : printed $\varepsilon$ OK at small $B$ ; $\hat{\sigma}$ still $\sim 10^4$ – $10^5$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>C31</b> | Incomplete public-MS gap at $N=1024$ : uncoveredBits= 10; approx candidate = .crypto.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>C32</b> | Track A approx: .crypto inject $B=1$ at $N=1024$ decodable; $\varepsilon \log_2 \approx -8.4$ at 8 trials (not $-64$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>C33</b> | Track A Metal SING covering .crypto at $N=1024$ with $B=1$ PASS (secret + public-ms).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>C34</b> | Track A covering baseLog= 4: $\varepsilon \log_2 \approx -913$ ( $\leq 2^{-64}$ ) + Metal SING PASS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>C35</b> | Track A covering baseLog= 2: $\varepsilon \leq 2^{-64}$ through inject $B=16$ ( $\varepsilon \log_2 \approx -65.4$ ) + Metal SING PASS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>C36</b> | Track A covering baseLog= 1: $\varepsilon \leq 2^{-64}$ through inject $B=32$ ( $\varepsilon \log_2 = -97.8$ , 95% bound $-71.2$ at $n=64$ ; the $-139$ was a 4-trial low draw) + Metal SING PASS; $B=128$ still fails.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>C37</b> | Track A Gaussian BK inject on covering-b1: $\sigma=24$ meets $\varepsilon \leq 2^{-64}$ (95% bound $-81.4$ at $n=20$ ) + Metal SING PASS; torus $\sigma \approx 128$ undecodable at native $k=1$ ( <b>C55</b> ); torus-scale close is <b>C52</b> ( $k=7$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>C38</b> | Encrypted sequential DFF host-clock: 4-bit counter Metal $N=1024$ SING PASS (15.88 s/4). Not PicoRV32.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>C39</b> | Encrypted historical one-byte / one-round legacy-UKW E256 teaching cone Metal $N=1024$ SING PASS (22.77 s/2). It predates live fixture-v4 and does not exercise transported centerMask/absoluteByteCounter, the conjugated-XOR center, live BRAM, NLFF, or the full core.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>C40</b> | Encrypted toy NOP/ADD ISA Metal $N=1024$ SING PASS (20.31 s/4). Not PicoRV32.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>C41</b> | Covering-b1 Gaussian $\sigma=128$ : meets $\varepsilon \leq 2^{-64}$ at $N=256$ (95% bound $-594.5$ ). At $N=512$ native $k=1$ is decodable but $\varepsilon \log_2 = -50.6$ , short of the bar; stride $k \geq 2$ recovers it (bound $-151.5$ at $k=2$ ) with Metal SING PASS at $k=2, 3, 7$ . Native $k=1$ at $N=1024$ still fails ( <b>C37</b> ); torus-scale close is <b>C52</b> .                                                                                                                                                                                                                                                                                                                |
| <b>C42</b> | 4-bit CSA vs ripple: LUT2 $11 \rightarrow 8$ ( $-27\%$ ) + encrypted SING PASS. Not TensorLUT melt.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

*Continued on next page*

| ID         | Result ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C43</b> | $k\delta$ encoding at $N=1024$ covering-b1 $\sigma=128$ : $k=4$ Metal SING PASS but $\varepsilon \log_2 = -34.1$ settled at $n=32$ , short of the bar. $k=8$ meets <i>both</i> — $\varepsilon$ bound $-113.6$ and SING PASS; its recorded SING FAIL predated the 2026-08-15 determinism fix.                                                                                                                                                     |
| <b>C44</b> | TensorLUT melt-freeze-snap on a separable Boolean interpolant (three lemmas). Not arbitrary-netlist melt.                                                                                                                                                                                                                                                                                                                                        |
| <b>C45</b> | Encrypted PicoRV32 1-tick SING at demo $N=8$ : 76.12 ms/row; 4-bit hardness. Not production $N$ .                                                                                                                                                                                                                                                                                                                                                |
| <b>C46</b> | Encrypted PicoRV32 10-tick <code>resetn</code> boot SING at $N=8$ : median 0.473 s/10 over five runs, 0.7% spread; $Q \equiv$ clear. The archived 0.972 s/10 predates the sequential wavefront and is stale. Plain <code>\$_DFF_P_</code> native- $D$ fix.                                                                                                                                                                                       |
| <b>C47</b> | Encrypted PicoRV32 NOP-fetch: 10 instruction fetches, PC 0+4 $k$ through 0 $x24$ ; 3.15 s/32.                                                                                                                                                                                                                                                                                                                                                    |
| <b>C48</b> | 2-LUT cascade melt-snap-emit $(a \wedge b) \oplus c$ ; 8-corner SING of snapped INIT (dual interpolant OK).                                                                                                                                                                                                                                                                                                                                      |
| <b>C49</b> | Encrypted PicoRV32 <code>addi+sw</code> : store <code>wdata=1</code> , RAM0=1; 3.79 s/48 @ demo $N=8$ . Load-back is <b>C50</b> .                                                                                                                                                                                                                                                                                                                |
| <b>C50</b> | Encrypted PicoRV32 <code>lw</code> sees stored 1 (NBA xfer); 3.77 s/48 @ demo $N=8$ . Not Metal PicoRV.                                                                                                                                                                                                                                                                                                                                          |
| <b>C51</b> | Metal PicoRV32 NOP-fetch @ demo $N=8$ : tiled BR, 2 fetches, 63.55 s/8. Not fused netlist / not $N=1024$ .                                                                                                                                                                                                                                                                                                                                       |
| <b>C52</b> | Stride- $k$ public-MS covering-b1 @ $N=1024$ $\sigma=128$ $k=7$ : $\varepsilon \log_2 = -132.5$ (95% bound $-83.2$ at $n=32$ ; the $-170$ at 8 trials was unsupported) + Metal secret/public-ms SING PASS. Native $k=1$ still <b>C37</b> .                                                                                                                                                                                                       |
| <b>C53</b> | Sequential covering-b1 counter @ $N=1024$ $\sigma=128$ $k=7$ : Metal secret/public-ms SING PASS. Not PicoRV.                                                                                                                                                                                                                                                                                                                                     |
| <b>C54</b> | Sequential covering-b1 toy ISA @ $N=1024$ $\sigma=128$ $k=7$ : Metal secret/public-ms SING PASS. Not PicoRV.                                                                                                                                                                                                                                                                                                                                     |
| <b>C55</b> | Native $k=1$ covering-b1 $\sigma=128$ LWE- $n$ map @ poly $N=1024$ : decodable at $n \leq 512$ but $\varepsilon > -64$ ; $n=1024$ undecodable ( <b>C37</b> ).                                                                                                                                                                                                                                                                                    |
| <b>C56</b> | <code>cryptoPublicMS</code> $B=1$ $k=7$ : identity decodable $\varepsilon \approx -12.6$ ; Metal public-ms SING FAIL. Torus $\sigma=128$ still undecodable.                                                                                                                                                                                                                                                                                      |
| <b>C57</b> | Covering-b2 ( $\ell=16$ ) at $N=n=1024$ , $\sigma=128$ : historical $k=7$ is short of the $-64$ bar. The integrated preregistered $k=14$ , 192-trial run gives a 24-event bound of $-93.8005$ and then full-adder SING <b>8/8 PASS</b> (48.1757 s evaluation; 969.48 s end to end). Exact noiseless primary inputs and Gaussian statistical circuit scope—not a hard BK bound, fresh-input-noise result, or arbitrary-netlist/depth certificate. |
| <b>C58</b> | PicoRV32 <code>abc -lut 6</code> : 2006 LUT ( $-58\%$ ) + CPU boolean SING PASS @ $N=64$ (1.35 s). Not covering / not $N=1024$ .                                                                                                                                                                                                                                                                                                                 |
| <b>C59</b> | Sequential wavefront: PicoRV lut6 $N=64$ CPU boolean 0.165 s/1 ( $\sim 8.2\times$ vs <b>C58</b> ).                                                                                                                                                                                                                                                                                                                                               |
| <b>C60</b> | PicoRV lut6 covering-b2 $k=7$ @ $N=1024$ : Metal 1-tick Q SING FAIL ( $\sim 33$ min, DFF want= 0 got= 1). Confirmed <i>structural</i> by a post-fix re-run failing at the identical cell.                                                                                                                                                                                                                                                        |
| <b>C61</b> | PicoRV lut6 covering-b1 $k=7$ @ $N=1024$ : Metal 1-tick Q SING FAIL ( $\sim 61$ min). Confirmed structural post-fix, same cell.                                                                                                                                                                                                                                                                                                                  |
| <b>C62</b> | Metal PicoRV lut6 $N=1024$ $e=0$ 1-tick Q SING PASS (373.89 s). Not covering.                                                                                                                                                                                                                                                                                                                                                                    |
| <b>C63</b> | PicoRV lut6 covering-b2 $\sigma=128$ @ $N=64$ : Q SING PASS ( $\sim 1.72$ s). Native $k$ . $k=7$ SIGTRAPs.                                                                                                                                                                                                                                                                                                                                       |

*Continued on next page*

| ID         | Result ( <i>continued</i> )                                                                                                                               |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C64</b> | Extract→KS $n=64$ at poly $N=1024$ : native- $k$ covering-b2 adder+counter SING. Not <b>LWE-176</b> .                                                     |
| <b>C65</b> | PicoRV lut6 covering Metal $N=1024$ extract→KS $n=64$ : Q SING PASS (b2 114s, b1 212s).                                                                   |
| <b>C66</b> | PicoRV lut6 covering 10-tick resetn boot extract→KS $n=64$ : Metal PASS 1136s/10.                                                                         |
| <b>C67</b> | Covering KS ladder: $n=128$ adder PASS; $n=256/512$ SIGTRAP.                                                                                              |
| <b>C68</b> | PicoRV lut6 covering NOP-fetch extract→KS $n=64$ : Metal PASS 911s/8, fetches $0x0, 0x4$ .                                                                |
| <b>C69</b> | Covering KS adder SING PASS at $n=256$ and $n=512$ . The $n=512$ FAIL was withdrawn 2026-08-15 (determinism artifact). <b>C67</b> SIGTRAP was identity×4. |

## B.2 Open hedges

| ID        | Asterisk                                                                                                                                                                                          |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>H1</b> | <b>C23</b> filled JSON. Production $ \Delta  = 4.5$ ; divergences are core-SVP vs Cost <b>rop</b> .                                                                                               |
| <b>H2</b> | <b>Closed</b> 2026-08-12 ( $Z_{2N}$ pack). Kept as history.                                                                                                                                       |
| <b>H3</b> | <b>C20/C21</b> SING bars met; serial NTT SING still loses vs persist.                                                                                                                             |
| <b>H4</b> | <b>C64–C66/C68</b> extract→KS $n=64$ . <b>C69</b> covering SING at $n=256$ and $n=512$ (the $n=512$ FAIL was withdrawn). <b>C67</b> trap was identity×4. Native $k=1$ at $n=N$ still <b>C37</b> . |
| <b>H5</b> | *PublicMS gadgets ( $g_0 = \delta$ ): on-lattice intent, not a closer of old <b>H2</b> .                                                                                                          |
| <b>H6</b> | TensorLUT / quarantine vs campaign: parallel research, not P1030680 PT.                                                                                                                           |
| <b>H7</b> | Catalog / Regenbogen / UEBUNG: negatives graded; middle ring $\neq A$ tested under VIII-fast (24/24 dead; 294 WO open); catalog parked @417, resume -bombe-from 418.                              |

## B.3 Standing non-implications

These were an unnumbered list until 2026-08-15. They are now **N** rows in the claim sheet, because the campaign report and the agent rules cite **N5**, **N6** and **N7** by number and those references had no resolvable target. Numbering follows the original clause order, so existing citations keep their meaning.

| ID        | Do not let prose imply                                                        |
|-----------|-------------------------------------------------------------------------------|
| <b>N1</b> | A new lattice assumption.                                                     |
| <b>N2</b> | “We invented TFHE” — prior art is Chillotti et al., Ducas–Micciancio.         |
| <b>N3</b> | Production keys are safe without an estimator run ( <b>H1</b> , <b>C23</b> ). |
| <b>N4</b> | Mock or trivial torus encoding = FHE ( <b>C1</b> is a shape laboratory).      |
| <b>N5</b> | P1030680 is decrypted ( <b>H6</b> ).                                          |
| <b>N6</b> | Campaign fitness = encrypted tick rate. Fitness is cleartext Metal batch.     |
| <b>N7</b> | TensorLUT = a U-534 / Thetis break ( <b>H6</b> ; <b>C19</b> is structural).   |
| <b>N8</b> | Side channels are measured. GPU power and timing are parked.                  |
| <b>N9</b> | Quantum attacks are analyzed. Not in scope anywhere in this corpus.           |

Wild ideas in Chapter 15 are trajectory, not results.

# Appendix C

## Fifteen-week syllabus

Suggested title: *Reconfigurable Homomorphic Computing*. 3 credits. Mix of lecture, seminar, and lab (Chapter 2). This table is a *future* outline, not a schedule you can run. The book is not ready to teach.

| Wk | Part | Classroom                                                                  |
|----|------|----------------------------------------------------------------------------|
| 1  | I    | Subject definition; living boxes; claim audit assigned.                    |
| 2  | II   | Netlists, Yosys $\$1ut$ , $B$ vs $N$ ; Lab 1.                              |
| 3  | II   | Torus, $LWE$ , $PBS$ sketch; $N = 4$ Toeplitz.                             |
| 4  | III  | Pillar I pipeline; oracle vs encrypted; Lab 2.                             |
| 5  | III  | $SING$ ; $Z_{2N}$ pack as a worked bug; Lab 3.                             |
| 6  | III  | Metal compiler Phase 1/2; fused $DNF$ as a case.                           |
| 7  | III  | Certificates; hardness table; Lab 4.                                       |
| 8  | —    | Midterm (foundations + five-cell); Lab 5 optional.                         |
| 9  | IV   | TensorLUT setup; Theorem 1 ( <b>C19</b> ); Lab 6.                          |
| 10 | IV   | Shatter vs hold; involution sandwich; campaign as case study (no decrypt). |
| 11 | V    | E256 contract; Red/Blue; fail-closed.                                      |
| 12 | VI   | Application gallery; student five-cell cards; C52–C54 vs remainders.       |
| 13 | VI   | Labs catch-up; claim-audit drafts due.                                     |
| 14 | VI   | Open hedges; pick-a-next-experiment.                                       |
| 15 | VI   | Frontier seminars; essays due; living-epoch recap.                         |

Final: take-home claim audit plus a short oral defense of one reproduced box and one hedge box.

The thing being assessed is whether a hedge survives contact with a question. Stating **H1** and then, under follow-up, sliding into “so it is 176-bit secure” is the characteristic failure mode, and noticing it in your own sentence is the skill the course is trying to build. Treat it as the main rubric line and as something to discuss in the viva — not as an automatic fail. Everyone does it once; the difference between a student who has learned the material and one who has not is whether they catch themselves.

## Appendix D

# Reproduce commands (course subset)

Canonical file: REPRODUCE.md. Cookbook: directives/parameter-cookbook.md. Build:

```
swift build -c release --product helut
make hardware-check # ownership + root compatibility integrity
```

Authored HDL is canonical under **Hardware/**; checked-in generated receipts are under **Generated/**; disposable regeneration belongs under **build/hardware/**. Root HDL and netlist names are compatibility entry points, not generation destinations.

### D.1 Documents

```
make textbook # this book
make paper # three-pillar report
make writeup # P1030680 campaign report
```

### D.2 Hardness (C23, H1)

```
.build/release/helut --hardness-table
./Scripts/helut_sage_estimate.sh
```

Native SageMath 10.9 at ~/Applications/SageMath-10-9.app. Production row: HELUT 175.7 vs estimator 180.2. Four of eight anchors exceed the 16-bit merge tolerance — do not quote 176 as estimator cost on every row.

### D.3 Encrypted SING (C4–C6, C38–C40)

```
.build/release/helut --bench Generated/Netlists/Examples/netlist.json --degree 8 \
  --bench-encrypted --cpu-only --sing --vectors 8
./Scripts/helut_encrypted_sing.sh
.build/release/helut --bench Generated/Netlists/Examples/counter_netlist.json --degree 8 \
  --bench-encrypted --cpu-only --sing --vectors 8
.build/release/helut --bench Generated/Netlists/Enigma256/e256_round1_netlist.json --
  degree 16 \
```

```
--bench-encrypted --cpu-only --sing --vectors 32
.build/release/helut --bench Generated/Netlists/Examples/toy_isa_netlist.json --degree 8 \
\
--bench-encrypted --cpu-only --sing --vectors 32
```

**C38** counter, **C39** frozen E256 round (LUT4,  $N \geq 16$ ), **C40** toy ISA. None of these is PicoRV32 or live BRAM SoftBus.

## D.4 Metal microbench and NTT (C18)

```
make test-metal-p1
.build/release/helut --bench-encrypted-micro --degree 1024 \
--trials 2 --warmup 1
```

Default at  $N > 64$  is tiled-kernel with inlined NTT. Do not assign `-metal-br-fused` at  $N = 1024$  as a timed lab.

## D.5 Wavefront SING (C20, C21)

Boolean 10.6s/8; crypto  $\ell = 2$  11.38s/8 at  $N = 1024$ . Logs already in `logs/` — students need not re-time Metal unless the instructor assigns a machine-local check.

```
.build/release/helut --bench Generated/Netlists/Examples/netlist.json --degree 1024 \
--bench-encrypted --sing --vectors 8 \
--paths 'blind-rotate-metal public-ms boolean'
.build/release/helut --bench Generated/Netlists/Examples/netlist.json --degree 1024 \
--bench-encrypted --sing --vectors 8 \
--paths 'blind-rotate-metal public-ms crypto'
```

## D.6 Measured noisy BK (C22)

Covering gadget only.  $\ell = 1$  `booleanPublicMS` cannot carry BK noise.

```
.build/release/helut --measure-bk-noise --degree 8 \
--trials 8 --bk-noise 64
.build/release/helut --measure-bk-noise --degree 128 \
--trials 4 --bk-noise 64
```

## D.7 Product-shaped $N=1024$ residual (C26)

Graded negative for `cryptoPublicMS` at  $N=1024$  (**H4** remainder; Grade B covering Track A is **C52–C54**). Expect undecodable inject  $B=64$ ;  $B=4$  on `.crypto` may pass  $\infty$ -norm with  $\varepsilon \log_2 \approx -1$  (not  $-64$ ). **C56**:  $k=7$  does not make Metal public-ms SING pass.

```
.build/release/helut --measure-bk-noise --degree 1024 \
--trials 2 --bk-noise 64
.build/release/helut --measure-bk-noise --degree 1024 \
--trials 2 --bk-noise 4
```

## D.8 Exact public-MS covering (C27)

```
swift test -c release --filter testGGSWPublicMSCoveringCertificate
```

Exact degrees among  $\{8, \dots, 2048\}$ :  $\{8, 128\}$  only — for any power-of-two word  $w$  (C29), not just  $q=2^{32}$ . Statement: directives/ggsw-public-ms-covering.md.

## D.9 Track B $\varepsilon$ vs inject $B$ (C30)

```
for B in 1 2 4 8 16 32 64; do
  .build/release/helut --measure-bk-noise --degree 128 \
    --trials 8 --bk-noise $B
done
```

Residual amplification stays large; printed  $\varepsilon \log_2 \approx -24$  only at  $B=64$ .

## D.10 Incomplete public-MS gap / Track A approx (C31/C32)

```
swift test -c release --filter testGGSWIncompleteCoveringCertificate
.build/release/helut --measure-bk-noise --degree 1024 \
  --trials 4 --bk-noise 1
```

$\text{uncoveredBits}(1024) = 10$ .  $\text{.crypto } B=1 \Rightarrow \varepsilon \log_2 \approx -8.4$  (8 trials). Statement: directives/ggsw-incomplete.

## D.11 Track A Metal covering-crypto SING (C33)

```
.build/release/helut --bench Generated/Netlists/Examples/netlist.json --degree 1024 \
  --bench-encrypted --sing --vectors 2 --bk-noise 1 \
  --paths 'blind-rotate-metal secret crypto'
.build/release/helut --bench Generated/Netlists/Examples/netlist.json --degree 1024 \
  --bench-encrypted --sing --vectors 2 --bk-noise 1 \
  --paths covering-crypto
```

Expect PASS; covering  $\text{.crypto}$ , not  $\text{cryptoPublicMS}$ .

## D.12 Track A covering-b4 $\varepsilon \leq 2^{-64}$ (C34)

```
.build/release/helut --measure-bk-noise --degree 1024 \
  --trials 8 --bk-noise 1 --covering-base-log 4
.build/release/helut --bench Generated/Netlists/Examples/netlist.json --degree 1024 \
  --bench-encrypted --sing --vectors 2 --bk-noise 1 \
  --paths covering-b4
```

Expect  $\varepsilon \log_2 \approx -913$ ; Metal PASS. Torus-scale  $B=128$  still fails.

### D.13 Track A covering-b2 $\varepsilon \leq 2^{-64}$ through $B=16$ (C35)

```
.build/release/helut-bench --measure-bk-noise --degree 1024 \  
  --trials 4 --bk-noise 16 --covering-base-log 2  
.build/release/helut-bench --bench Generated/Netlists/Examples/netlist.json --degree 1024 \  
  --bench-encrypted --sing --vectors 2 --bk-noise 16 \  
  --paths covering-b2
```

Expect  $\varepsilon \log_2 \approx -65.4$  at  $B=16$ ; Metal PASS.  $B=32$  fails the bar.

### D.14 Track A covering-b1 $\varepsilon \leq 2^{-64}$ through $B=32$ (C36)

```
.build/release/helut-bench --measure-bk-noise --degree 1024 \  
  --trials 4 --bk-noise 32 --covering-base-log 1  
.build/release/helut-bench --bench Generated/Netlists/Examples/netlist.json --degree 1024 \  
  --bench-encrypted --sing --vectors 2 --bk-noise 32 \  
  --paths covering-b1
```

Expect  $\varepsilon \log_2 \approx -139$  at  $B=32$ ; Metal PASS.  $B=128$  still fails.

### D.15 Track A $\sigma=128$ covering-b1 at $N \leq 512$ (C41)

```
.build/release/helut --measure-bk-noise --degree 512 --trials 4 \  
  --bk-noise-sigma 128 --covering-base-log 1  
.build/release/helut --bench Generated/Netlists/Examples/netlist.json --degree 512 \  
  --bench-encrypted --sing --vectors 2 --bk-noise-sigma 128 \  
  --paths covering-b1
```

Does not close  $N=1024$  at native  $k=1$ . Headroom is  $\delta = q/(2N)$ .

### D.16 $k\delta$ encoding at $N=1024$ (C43)

```
.build/release/helut --measure-bk-noise --degree 1024 --trials 4 \  
  --bk-noise-sigma 128 --covering-base-log 1 --boolean-scale-mul 4  
.build/release/helut --bench Generated/Netlists/Examples/netlist.json --degree 1024 \  
  --bench-encrypted --sing --vectors 2 --bk-noise-sigma 128 \  
  --paths covering-b1 --boolean-scale-mul 4
```

$k=4$  Metal SING PASS; 4-trial  $\varepsilon \log_2 \approx -43$  (not  $-64$ ).  $k=8$  meets  $\varepsilon$ ; public-ms SING fails under  $/k\delta$  refresh.

### D.17 Stride- $k$ public-MS covering-b1 (C52)

```
.build/release/helut --measure-bk-noise --degree 1024 --trials 8 \  
  --bk-noise-sigma 128 --covering-base-log 1 --boolean-scale-mul 7  
.build/release/helut --bench Generated/Netlists/Examples/netlist.json --degree 1024 \  
  --paths covering-b1
```

```
--bench-encrypted --sing --vectors 2 --bk-noise-sigma 128 \
--paths covering-b1 --boolean-scale-mul 7
```

Expect 8-trial  $\varepsilon \log_2 \approx -170$ ; Metal secret + public-ms PASS.

## D.18 Sequential covering counter (C53)

```
.build/release/helut --bench Generated/Netlists/Examples/counter_netlist.json --degree
1024 \
--bench-encrypted --sing --vectors 2 --bk-noise-sigma 128 \
--paths covering-b1 --boolean-scale-mul 7
```

Same encoding as C52. Expect Metal secret + public-ms PASS. Not PicoRV.

## D.19 Sequential covering toy ISA (C54)

```
.build/release/helut --bench Generated/Netlists/Examples/toy_isa_netlist.json --degree
1024 \
--bench-encrypted --sing --vectors 2 --bk-noise-sigma 128 \
--paths covering-b1 --boolean-scale-mul 7
```

Expect Metal secret + public-ms PASS. Not PicoRV.

## D.20 Native $k=1$ LWE- $n$ map (C55)

```
.build/release/helut --measure-bk-noise --degree 1024 --trials 8 \
--bk-noise-sigma 128 --covering-base-log 1 --lwe-dimension 256
```

Omit `-lwe-dimension` for  $n=N=1024$  (C37). Expect decodable  $\varepsilon > -64$  at  $n \leq 512$ ; undecodable at  $n=1024$ . None  $\leq -64$ .

## D.21 cryptoPublicMS plus $k=7$ (C56)

```
.build/release/helut --measure-bk-noise --degree 1024 --trials 4 \
--bk-noise 1 --boolean-scale-mul 7
.build/release/helut --bench Generated/Netlists/Examples/netlist.json --degree 1024 \
--bench-encrypted --sing --vectors 2 --bk-noise 1 \
--paths 'blind-rotate-metal public-ms crypto' \
--boolean-scale-mul 7
```

Expect identity decodable at  $B=1$  ( $\varepsilon \approx -12.6$ ); Metal public-ms SING FAIL. Torus  $\sigma=128$  still undecodable.

## D.22 Covering-b2 cheaper SING (C57)

```
.build/release/helut --measure-bk-noise --degree 1024 --trials 4 \
  --bk-noise-sigma 128 --covering-base-log 2 --boolean-scale-mul 7
.build/release/helut --bench Generated/Netlists/Examples/regex_netlist.json --degree 1024 \
  --bench-encrypted --sing --vectors 1 --bk-noise-sigma 128 \
  --paths 'public-ms covering-b2' --boolean-scale-mul 7
```

Expect  $\varepsilon \log_2 \approx -110.7$ ; regex Metal public-ms PASS. Covering-b4 public-ms and E256 covering-b2 FAIL.

## D.23 PicoRV abc -lut 6 (C58)

```
mkdir -p build/hardware/PicoRV32
yosys -p "read_verilog Hardware/RTL/Vendor/PicoRV32/picorv32.v; \
  synth -top picorv32 -flatten; abc -lut 6; check -assert; \
  write_json build/hardware/PicoRV32/picorv32_lut6_netlist.json"
.build/release/helut --bench build/hardware/PicoRV32/picorv32_lut6_netlist.json --degree
  64 \
  --bench-encrypted --cpu-only --sing --vectors 1 \
  --paths 'blind-rotate public-ms boolean'
```

Expect 2006 LUT, PASS  $\sim 1.35$  s/1, 32-bit.  $N=8$  traps. Wavefront (C59): same command  $\sim 0.165$  s/1.

## D.24 PicoRV covering-b2 Q SING FAIL (C60)

```
.build/release/helut --bench Generated/Netlists/PicoRV32/picorv32_lut6_netlist.json --
  degree 1024 \
  --bench-encrypted --sing --vectors 1 --bk-noise-sigma 128 \
  --paths 'public-ms covering-b2' --boolean-scale-mul 7
```

Expect  $\sim 33$  min then DFF Q SING FAIL.

## D.25 Track B Metal SING with noisy BK (C28)

```
.build/release/helut --bench Generated/Netlists/Examples/netlist.json --degree 128 \
  --bench-encrypted --sing --vectors 4 --bk-noise 64 \
  --paths 'blind-rotate-metal public-ms crypto'
```

Covering degree. Expect PASS and non-zero decodable  $B_{\text{bk}}$ .

## D.26 TensorLUT Theorem 1 (C19)

```
swift test -c release --filter testTensorLUTFormalCertificate
```

Six lemmas must hold. Statement: directives/tensorlut-theorem.md. Not a U-534 / P1030680 decrypt.

## D.27 Enigma256 SoftBus Theorem 2 (C24)

```
swift test -c release --filter testEnigma256FormalCertificate
```

Five lemmas must hold. Statement: `directives/enigma256-theorem.md`. Structural SoftBus contract—not IND-CPA.

## D.28 TensorLUT Theorem 1 corollary (C25)

```
swift test -c release --filter testTensorLUTFormalCorollaryCertificate
```

Emitter-discrete agreement and involution-under-freeze must hold. Statement: `directives/tensorlut-theorem.m` (corollary).

## D.29 TensorLUT melt-freeze-snap (C44)

```
swift test -c release --filter testTensorLUTMeltFreezeSnapCertificate
```

Three lemmas must hold (unique maximizer, snap basin, freeze). Statement: `directives/tensorlut-theorem.m` (Theorem 1''). Not multi-LUT topological melt.

## D.30 Encrypted PicoRV32 1-tick (C45)

```
.build/release/helut --bench Generated/Netlists/PicoRV32/picorv32_netlist.json --degree 8 \
--bench-encrypted --cpu-only --sing --vectors 1 \
--paths 'blind-rotate public-ms boolean' \
| tee logs/helut-encrypted-n8-cpu-sing-picorv32.log
```

Expect PASS, ~76 ms/row, hardness 4.0 bits (demo  $N$ ). Output SING;  $Q$  SING is **C46**.

## D.31 Encrypted PicoRV32 10-tick boot (C46)

```
.build/release/helut --bench Generated/Netlists/PicoRV32/picorv32_netlist.json --degree 8 \
--bench-encrypted --cpu-only --sing --ticks 10 --reset-hold 3 \
--paths 'blind-rotate public-ms boolean' \
| tee logs/helut-encrypted-n8-cpu-sing-picorv32-boot10.log
```

Expect PASS 0.972 s/10,  $Q \equiv$  clear. Idle memory. LUT-tax:  $N=32 \sim 819$  ms;  $N=64 \sim 4.35$  s.

## D.32 Encrypted PicoRV32 NOP-fetch (C47)

```
.build/release/helut --bench Generated/Netlists/PicoRV32/picorv32_netlist.json --degree 8 \
--bench-encrypted --cpu-only --sing --ticks 32 --reset-hold 3 \
--encrypted-mem nop --paths 'blind-rotate public-ms boolean' \
| tee logs/helut-encrypted-n8-cpu-sing-picorv32-nop-fetch.log
```

Expect 10 fetches, addresses  $0, 4, \dots, 0x24$ ,  $Q \equiv \text{clear}$ .

### D.33 Encrypted PicoRV32 addi+sw (C49)

```
.build/release/helut --bench Generated/Netlists/PicoRV32/picorv32_netlist.json --degree 8 \
--bench-encrypted --cpu-only --sing --ticks 48 --reset-hold 3 \
--encrypted-mem prog --paths 'blind-rotate public-ms boolean' \
| tee logs/helut-encrypted-n8-cpu-sing-picorv32-prog-store.log
```

Expect STORE wdata=1, RAM0=1, 14 fetches,  $\sim 79$  ms/row.

### D.34 Encrypted PicoRV32 lw sees 1 (C50)

```
.build/release/helut --bench Generated/Netlists/PicoRV32/picorv32_netlist.json --degree 8 \
--bench-encrypted --cpu-only --sing --ticks 48 --reset-hold 3 \
--encrypted-mem prog --paths 'blind-rotate public-ms boolean' \
| tee logs/helut-encrypted-n8-cpu-sing-picorv32-prog-lw.log
```

Expect LOAD xfer rdata=1, STORE wdata=1, 14 fetches,  $\sim 79$  ms/row,  $Q \equiv \text{clear}$ . Demo  $N=8$ .

### D.35 Metal PicoRV32 NOP-fetch (C51)

```
.build/release/helut --bench Generated/Netlists/PicoRV32/picorv32_netlist.json --degree 8 \
--bench-encrypted --sing --ticks 8 --reset-hold 3 \
--encrypted-mem nop --paths 'blind-rotate-metal public-ms boolean' \
--metal-br-tile 8 \
| tee logs/helut-encrypted-n8-metal-sing-picorv32-nop-fetch.log
```

Expect 2 fetches, PASS  $\sim 64$  s/8. Use tiled BR; fused traps.

### D.36 2-LUT cascade melt+emit (C48)

```
swift test -c release --filter testTwoLUTCascadeMeltFreezeSnapEmit
```

Eight Boolean corners of the snapped INIT must match  $(a \wedge b) \oplus c$ .

### D.37 Boolean oracle benches (C1)

```
./Scripts/helut_boolean_bench.sh
./Scripts/helut_boolean_scale.sh
```

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